ARTICLE 16 October 2026

# How autonomous research shapes scientific understanding

Jiangjie Qiu[1,*], Yijun Li[1,*], Yaotian Yang[1,*], Honghao Chen[1], Wentao Li[1], Xiaonan Wang[1,†]
[1]Beijing Key Laboratory of Artificial Intelligence for Advanced Chemical Engineering Materials, State Key Laboratory of Chemical Engineering and Low-Carbon Technology, Department of Chemical Engineering, Tsinghua University, Beijing 100084, China.
* Jiangjie Qiu, Yijun Li and Yaotian Yang contributed equally.
† Correspondence: wangxiaonan@tsinghua.edu.cn.

All figures are ImageGen previews with illustrative geometry; exact numerical evidence is retained separately.

**As AI scientists and autonomous laboratories move towards sustained research cycles, machines are evolving from experimental executors into autonomous organizers of scientific inquiry. Yet how scientific understanding emerges within autonomous research and guides its continuation remains poorly understood. Here we introduce ChemWorld, programmable chemical worlds governed by thermodynamic, kinetic and coupled physicochemical laws. Agents autonomously organize experiments, acquire evidence and adapt exploration, while scientific laws and information conditions remain independently controllable. Through repeated investigations and controlled comparisons across six chemical systems, we connect research goals, prior knowledge, experimental choices and evolving scientific judgments. Agents formulate and test expectations, revise explanations and establish useful scientific relationships that guide subsequent experiments. Different exploratory and inferential paths can yield similar predictive performance, whereas experimental achievement, local understanding and transfer do not always advance together: genuine local discoveries can coexist with departures from evidential support in subsequent generalization. These results reveal how knowledge generation and the organization of exploration shape one another: agents develop understanding through research and use that understanding to redirect further inquiry. This work advances the study of autonomous science from the realization of experimental cycles towards an empirical understanding of how machines generate scientific knowledge, organize further inquiry and sustain scientific discovery.**

An autonomous researcher does more than act on the world. By choosing experiments and measurements, it also shapes the experience from which its scientific judgments form: which phenomena become visible, which explanations encounter a test and which results guide the next investigation. As AI scientists and self-driving laboratories undertake increasingly complex experimental cycles[1,2], this organization of inquiry becomes part of the scientific activity delegated to machines. The promise of autonomous science therefore concerns how experimental agency becomes scientific understanding.

Robot scientists and autonomous laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[3–6]. Language-model agents extend the connection between scientific instructions, tools and laboratory operations[7,8]. These advances make a broader question experimentally accessible: how does scientific knowledge form within autonomous research, and how does developing knowledge reshape further inquiry? An effective procedure and an accurate prediction establish important outcomes, but neither alone reveals the path through which a judgment arose or what the research established[9,10].

Scientific understanding develops through observations and through the ideas used to interpret them. Einstein's 1905 paper opens with an asymmetry in descriptions of magnet–conductor motion, before developing new principles and their consequences for space and time[11]. Proposing explanatory hypotheses, or abduction, contributes to scientific understanding alongside deduction and empirical generalization[12]. Within autonomous inquiry, an explanation can suggest an experiment; the observation can support, narrow or displace that explanation; and the resulting judgment can guide another intervention[13–15]. The evidence available to the researcher is therefore itself an outcome of inquiry. A successful experiment can establish a useful local relationship, while an unsuccessful one can advance understanding by ruling out an expectation. Following these transitions connects the formation of knowledge to the exploration it enables.

We study these connections in ChemWorld. Programmable chemical worlds allow agents to choose materials, operations and measurements while an evaluator independently controls scientific laws and supplied information. Repeated investigations across six systems connect experimental choices to evolving judgments. Controlled comparisons examine research goals, prior information and access to experimentation; continuous records follow how relationships are established, revised and used. Together, these studies examine the organization of inquiry, the development of scientific understanding and the nature and conditions of progress in autonomous research.

## Making autonomous scientific inquiry experimentally tractable

ChemWorld combines materials, operations and instruments into stateful chemical worlds governed by thermodynamic, kinetic and coupled process relationships. Reaction, thermal processing, equilibrium, separation and crystallization components connect into multistage procedures. Sample states persist: an upstream reaction or thermal history can change a later separation or measurement. Agents choose quantities, processing sequences, measurements and subsequent experiments within fixed resource allowances. An evaluator independently controls world parameters and public information, records committed operations and obtains withheld reference outcomes (Fig. 1).

This arrangement connects the selection of evidence to its interpretation. Knowledge tests and supplied-data tasks assess important scientific abilities[16,17], and interactive discovery environments already connect action to observation[18–21]. Recent agent studies also examine the revision of chemical structure hypotheses and the construction of explanatory representations[22,23]. ChemWorld follows inquiry through composable chemical processes. Public records identify what the agent saw and proposed; execution records establish which interventions changed the sample; independent world responses test later predictions. A requested operation can fail without producing the intended experiment, and a relationship supported in one setting may remain unresolved in another.

The design covers 450 planned investigations in four complementary matrices. The objective–information matrix contains ninety electrochemical and ninety reaction-processing sessions. The process matrix contains forty-five sessions each in partitioning, purification, equilibrium and reaction-coupled crystallization. The opportunity matrix compares research with no research in equilibrium and crystallization, thirty sessions per system. Thirty further investigations use matched aqueous worlds (Methods; Extended Data Fig. 2).

Opaque information supplies the task and operating contract without an additional instance dossier; Aligned supplies correct but incomplete information; MisIndexed supplies a specified mismatch. Discovery commissions emphasize investigating relationships, whereas optimization commissions emphasize developing an effective procedure. Worlds, interfaces and allowances are held to each fixed block's design, with three fresh sessions per cell and no controlled model sampling seed. These contrasts specify the conditions of research; the agents organize the actual paths.

After experimentation, researchers seal a scientific account, predict twelve new interventions without laboratory access and reflect without reference-truth feedback. We connect contemporaneous judgments to operations, observations, revisions and later use. All 300 public histories in the new experimental block were read, including incomplete histories and unsuccessful tests; illustrative trajectories were then selected for their explanatory content. Study labels connect the cases below to their full records and critical times in Supplementary Information. The 450-session design establishes coverage; condition comparisons estimate outcomes; selected histories establish particular transitions rather than their prevalence. Throughout, we distinguish world truth, the evidence available at a given time and the judgment adopted.

## The organization and adaptation of scientific inquiry

Research starts from goals, available information and a constrained action space. Agents choose which conditions to compare, which procedures to repeat and when to pursue a new question. These decisions determine both the evidence acquired and the opportunities still available. Goal and information contrasts show the consequences of different starting conditions; continuous records reveal how observations and judgments enter the adaptation of inquiry.

In the repeated electrochemical matrix, optimization rather than discovery increases world-equal mean retested recommendation score from 0.4813 to 0.5852 and product yield from 0.1310 to 0.4225. Prediction errors nevertheless increase for score (0.1539 to 0.1829), yield (0.0665 to 0.0781) and Faradaic efficiency (0.1677 to 0.2218), each in all five worlds (Fig. 2a,b). Developing a more effective procedure does not ensure better judgment of new interventions.

Information has similarly differentiated consequences. Under reaction optimization, Aligned rather than Opaque information improves recommendation score from 0.2735 to 0.4429 and yield from 0.5502 to 0.7116. Risk prediction error falls from 0.2389 to 0.1226, whereas conversion error rises from 0.0355 to 0.1595; all four directions occur in all five worlds (Fig. 2c,d). Thus the same information package can support more effective interventions and more accurate risk judgments while leaving conversion less accurately predicted.

Continuous records show how a result enters the next experimental choice. In purification study A, the researcher distinguishes the response it expects from the result required to continue a procedure (Fig. 3a). A no-wash procedure gives purity 0.8140 and recovery 0.0479. Increasing the organic-phase amount raises recovery but leaves purity at 0.7886, below the required 0.80. Before batch 11, the researcher predicts that a small 0.005 L wash will cross the purity threshold. It also sets a prospective action criterion: "if it passes with recovery above 0.0479, replicate it in batch 12." This separates an expected scientific response from the condition for pursuing the procedure.

Batch 11 reaches purity 0.8013 but recovery only 0.0441. The purity expectation is met, while the continuation criterion is not. Batch 12 therefore repeats the earlier no-wash procedure, yielding purity 0.8149 and recovery 0.0368. The final account recognizes repeated qualification alongside variation in recovery, and later predictions retain the observed purity–recovery trade-off. The two responses play different roles in the next decision: the wash meets the expected purity threshold, but its recovery determines whether the researcher pursues it.

Purification study B shows how a judgment changes the evidence still obtainable. After four low-recovery batches, the researcher proposes that a feed component was omitted and states that it will add that component in the remaining experiments. The next four batches implement both addition commands, which consume the same shared stock. Stock depletion then prevents the planned temperature experiments. In the final note, the researcher correctly excludes those rejected or blank operations from temperature-response evidence. Acting on an interpretation of the feed thus changes material use and the scope of subsequent inquiry.

Research paths can also differ within the same scientific problem. Historical five-model equilibrium campaigns illustrate two routes to accurate dilute-region predictions (Extended Data Fig. 7). An Astra researcher samples the dilute region, observes a departure from the higher-concentration

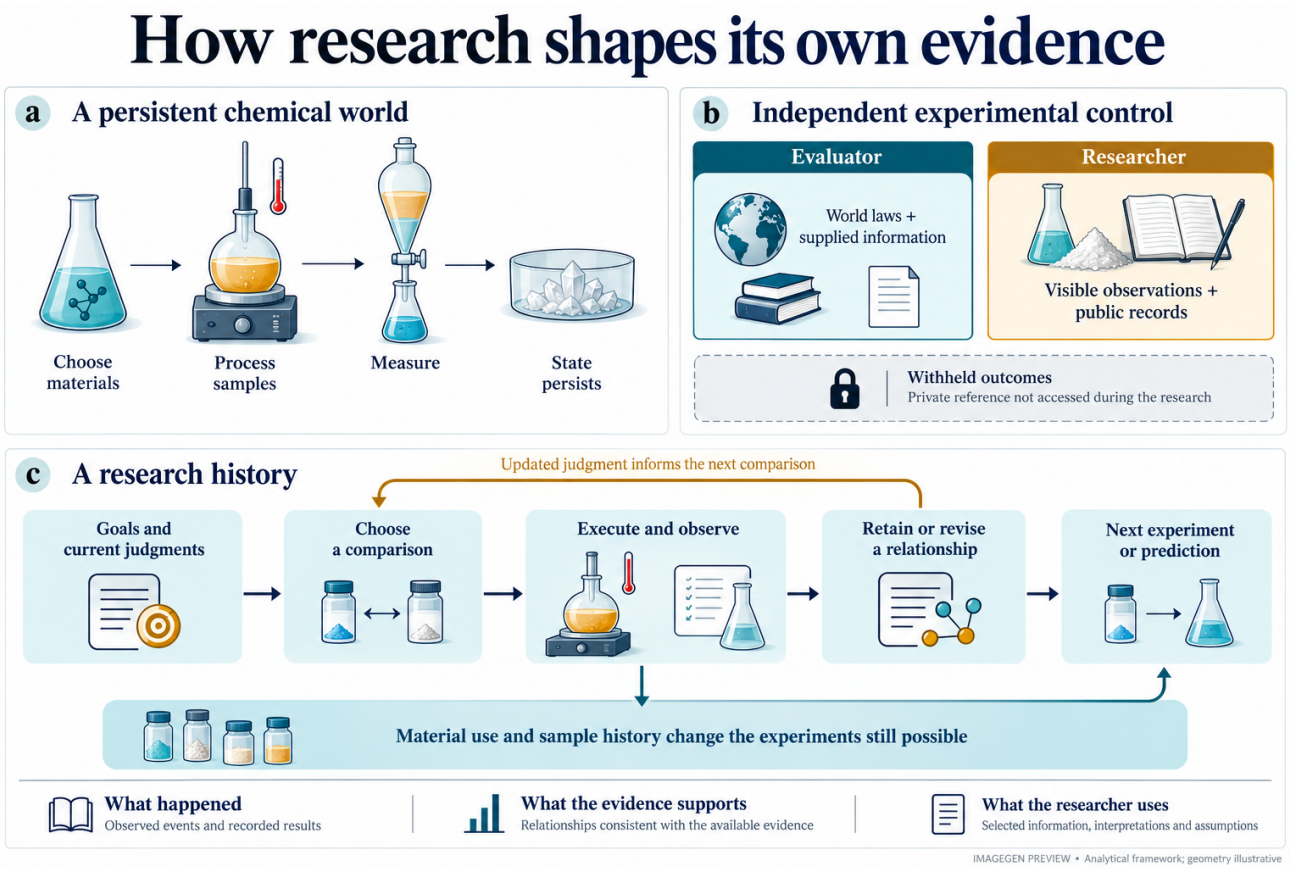



Figure 1 | Autonomous research shapes its own evidence. a, Agents choose materials, operations and measurements within a persistent chemical world. b, World laws and supplied information are independently controlled while reference outcomes remain withheld. c, Public expectations, chosen comparisons, observations, revised judgments and subsequent actions are connected through a research history. Resources and sample states constrain which experiments remain possible. ImageGen conceptual preview.

plateau and uses that change in its subsequent account and forecast. Sol/Opaque campaigns achieve accurate dilute predictions through extrapolation from local observations without directly sampling that region. Observing a new response and successfully inferring one are distinct scientific achievements. Their similar predictive outcomes make the histories of evidence and inference essential to understanding what each study contributes.

## The formation and evolution of scientific understanding

Scientific understanding develops as observations acquire meaning for further inquiry. A relationship can support a new test, an unexpected result can narrow an explanation, and an unresolved boundary can become an experimental question. The following histories connect these changes to comparisons that the researchers themselves propose and carry out.

Reaction study A follows an expectation from observation through a new experiment to later use (Fig. 3b). At 390 K and 1,800 s heating, direct termination gives conversion 0.41718. A later batch with the same heating followed by an unquenched 1,800 s wait reaches 0.61847 despite cooling by about 44 K. The researcher treats this as evidence that stopping external heating has not ended the chemical change. Before the final batch, it predicts that quenching before the same wait should preserve conversion near 0.42, rather than the approximately 0.62 unquenched-wait result.

The final batch implements that order and gives conversion 0.42724. The subsequent note retains a local relationship between the quench operation and suppression of further conversion. This evidence establishes which part of the processing history remains chemically consequential. Later forecasts use the relationship in new procedures, while transfer to unobserved solvent and loading conditions remains less accurate. The comparison supports the operational expectation under the tested sequence; its differing cooling histories leave thermal and other effects unresolved (Methods).

Electrochemical study B, commissioned to optimize, makes the scope of polarity dependence an experimental question (Fig. 3c). After ten batches of material and duration investigation, the researcher records that positive-polarity specificity has not undergone a matched reverse-sign test. It reserves that comparison and an exact repeat, writing: “If negative potential retains high yield, the assumed polarity specificity is rejected”. Batch 11 at −1.0 V retains yield 0.384; the positive recipe and its final repeat give approximately 0.911 and 0.905. The terminal account changes from apparent polarity specificity to polarity advantage, and subsequent forecasts retain reverse-polarity production. The revised relationship remains useful while

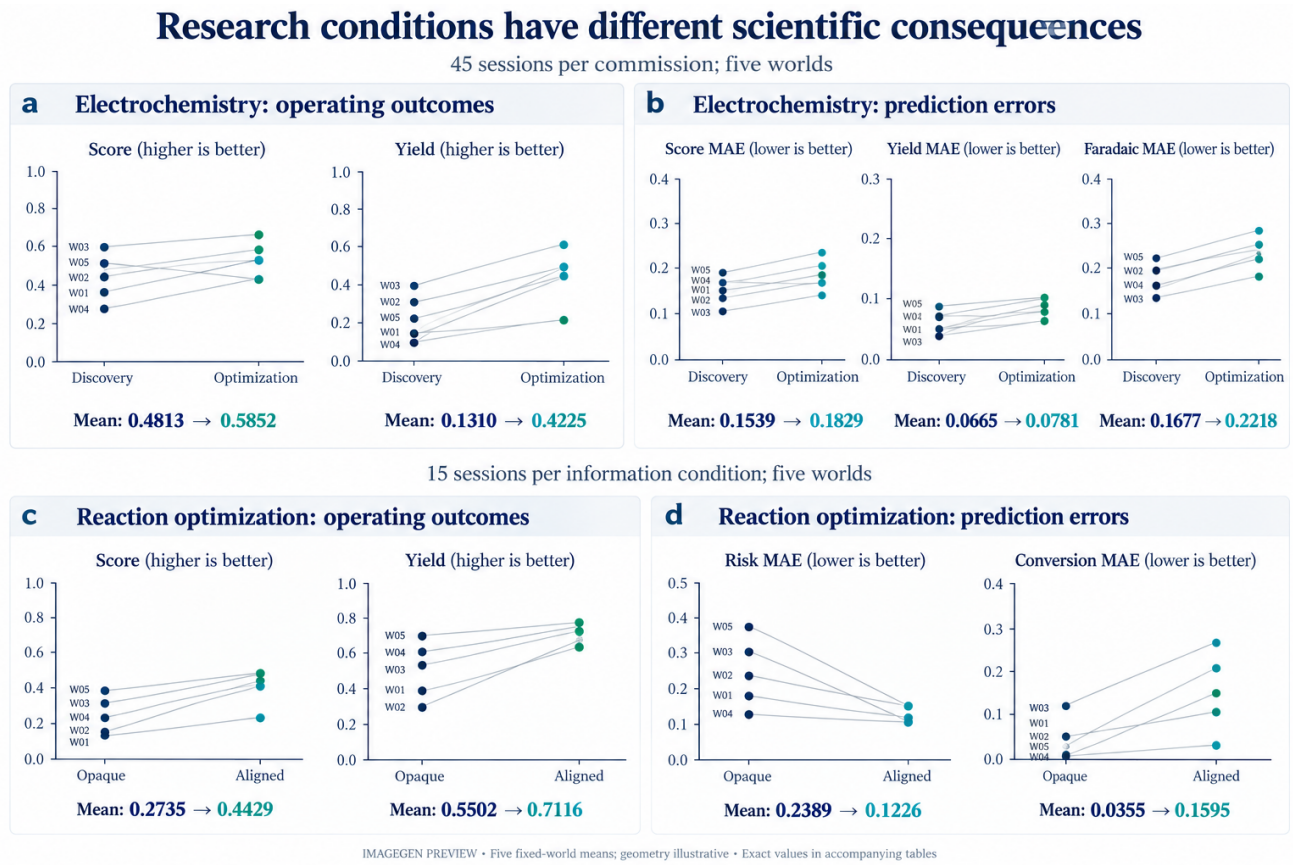



Figure 2 | Goals and information have different consequences for operation and prediction. a,b, Electrochemical discovery-optimization comparisons, forty-five sessions per commission across five fixed worlds and three information conditions. c,d, Reaction optimization under Opaque and Aligned information, fifteen sessions per arm. The preview illustrates within-world comparisons; displayed summary values refer to retained results. Generated dots and lines do not reproduce exact session values, world means or uncertainty. Exact values and available-response denominators are retained in Supplementary Information. ImageGen preview; geometry is illustrative.

becoming more precise: progress here consists in replacing apparent exclusivity with a supported advantage.

The three histories connect judgment to research in different ways. Purification applies an existing continuation criterion to a new observation; reaction processing tests a local operational expectation; electrochemistry revises the content of a scientific relationship. These developments change what the researcher has reason to pursue or expect. The value of a judgment lies partly in the new comparisons it opens and in how their results develop the relationship further.

Electrochemical study A provides a contrasting path from a local result to subsequent judgment. It begins with a matched sign comparison: negative potential produces a response where the tested positive setting is effectively inactive. Subsequent batches refine materials and duration on the negative side. The terminal note explicitly limits the sign result to the tested recipe. Yet later numerical forecasts assign near-background responses to untested positive potentials within the examined magnitude range, with intervals missing the references. A stated boundary and its application in new judgments can therefore diverge.

The matched-world matrix examines how different underlying chemistry enters the evidence and subsequent judgments (Fig. 4). Five pairs of aqueous worlds share public framing and selected parameters but differ in coupling to an additional neutral ion pair.

In matched-world study A, the researcher states that a sharp precipitation onset with a buffered dissolved response would support a solubility-limited interpretation, while smooth changes would also admit an empirical partition description. After six assays it proposes denser concentration sampling, a central repeat and increased loading; the next six batches implement those choices. The twelve actual observations provide the evidence behind the adopted relation (Fig. 4d; Supplementary Information). Its final account retains a smooth monotone relationship, and its forecast explicitly uses that relationship because no sharp threshold was resolved. Matched-world study B predicts a response to diluting a concentrated sample from 0.75 to 0.375 M and then performs the dilution. Normalized pH rises from 0.08045 to 0.10803. The researcher supports approximate reversibility, while the nearest separately prepared comparator is 0.4 M and the precipitation readings come from different instruments. These records connect stated alternatives, chosen comparisons and the relationships subsequently used.

Across the matrix, replaying chosen action sequences in their paired worlds shows that every one of thirty sessions acquires at least one observation differing by three assay standard deviations in two response channels. The chosen experiments therefore encounter physically distinguishable responses.

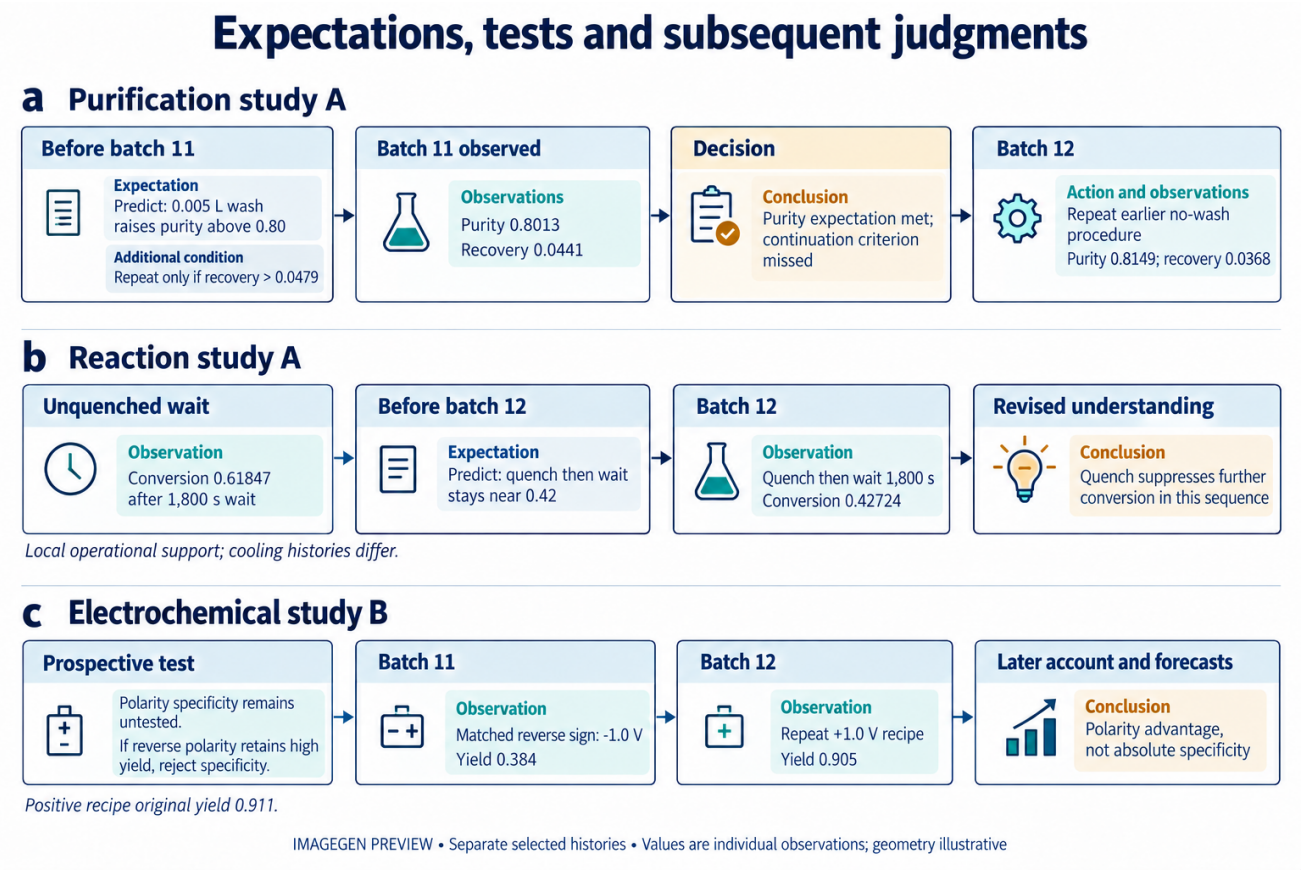



Figure 3 | Expectations, tests and subsequent judgments. a, Purification study A: the small-wash purity expectation is met, but recovery misses the prospective continuation criterion, prompting a repeat of the no-wash procedure. b, Reaction study A: an unquenched wait motivates a quench-before-wait prediction and locally supporting test. c, Electrochemical study B: the stated rejection criterion precedes a reverse-sign test; retained production changes specificity to polarity advantage. The final batch repeats the positive recipe. Rows are separate selected histories; values are individual observations. ImageGen preview; exact records and timing are in Supplementary Information.

Final forecasts also reflect these world differences. A post hoc analysis compares B-minus-A differences in the three-session forecast means with differences in the five-observation reference means at twelve common queries (Fig. 4a–c). Among nonzero reference differences, directions agree in 54/57 pH, 48/57 dissociation and 39/39 precipitation contrasts. The remaining 3, 3 and 21 reference differences are zero. All 180 contrasts remain in the analysis; in the descriptive subset reaching three assay standard deviations, directions agree in 48/48, 10/10 and 27/27 cases, respectively.

The forecasts capture difference magnitude as well as direction (Fig. 4e). Mean absolute contrast errors are 0.002256 for normalized pH, 0.002945 for dissociation and 0.002368 for precipitation. Giving identical forecasts in the two worlds would instead yield errors of 0.026007, 0.007239 and 0.024840. Forecast errors are thus 91.3%, 59.3% and 90.5% lower than this zero-difference reference, with an advantage in all five world pairs for each response. These are properties of the final three-session mean forecasts. The block has no no-research arm and supplies close but nonidentical starter observations, so it does not isolate a research-induced gain.

Complementary histories show that learning can also proceed through failed expectations (Extended Data Fig. 8). Stronger mixing defeats a provisional partitioning plateau; a fourth crystallization thermal cycle fails to extend the improvement from the third; further dilution removes recoverable particles and narrows an apparent operating window. These tests contribute through the judgments they revise, even when they do not improve the final procedure. The effects of simultaneously changed factors remain unresolved where the experiments did not separate them.

Interpreting an experimental history also requires determining which operations actually occurred. Some researchers treat rolled-back operations as completed tests, while others explicitly exclude them (Extended Data Fig. 6). In the focal heating case, the public return combines success labels with a rollback message and zero time and temperature changes. The resulting account incorporates this ambiguous execution history into later judgments. These cases expose how interface signals and their interpretation jointly shape the evidence a researcher treats as available.

Across these histories, a successful intervention can establish a bounded relationship, while a failed expectation can produce a warranted revision. An empirical regularity can guide further research even while its mechanism remains open. Understanding develops through changes in what is compared, what the evidence establishes and what remains worth investigating.

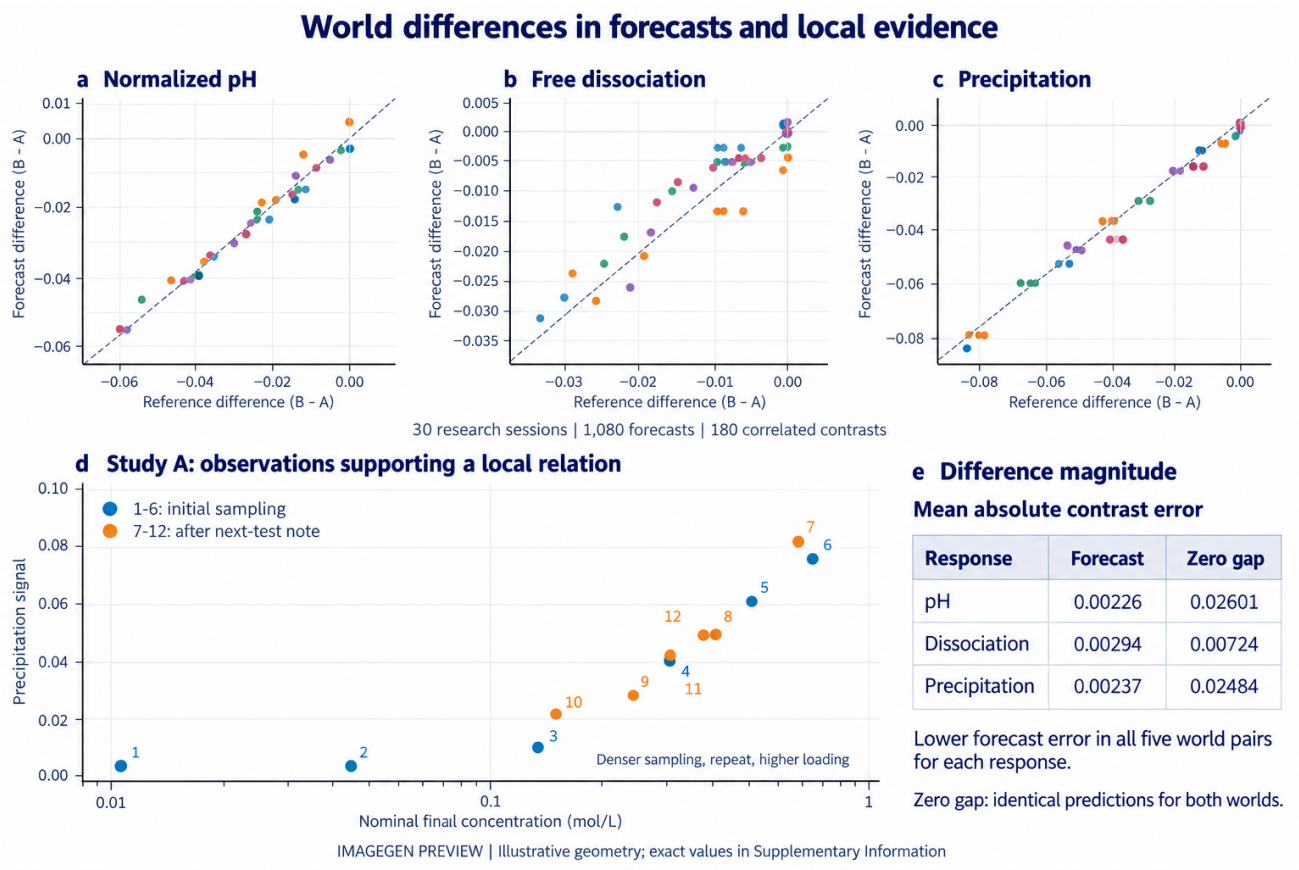



Figure 4 | World differences in forecasts and the observations supporting a local relationship. a–c, Reference B-minus-A responses and corresponding differences in three-session forecast means. Analysis retains all 180 correlated contrasts, including zero and small reference differences; references average five observations per query-world. Dashed lines denote equality. d, The twelve assays in matched-world study A; batches 7–12 follow the next-test note. e, Absolute contrast errors compared with the zero-difference reference: identical predictions in both worlds. Numerical labels report retained results. ImageGen preview; plotted positions are illustrative, with exact values and pair-level summaries in Supplementary Information.

## The nature and conditions of scientific progress

The preceding histories locate progress in establishing relationships, revising expectations and choosing further tests. We next examine what this research enables under new interventions. After experimentation ends, agents answer new questions without further laboratory access. The opportunity matrix compares their predictions with those of agents given no research opportunity. In each of two systems, five fixed worlds and three fresh sessions per arm compare twelve-batch research with no source experimentation under common Opaque information and assessment. The contrast estimates the complete opportunity to choose experiments, receive feedback and record judgments.

The crystallization comparison establishes substantial benefits (Fig. 5a). All thirty sessions provide valid forecasts. Relative to no research, world-equal mean absolute error falls from 0.13853 to 0.01661 for purity and from 0.23619 to 0.07641 for size index. Recovery error falls from 0.13532 to 0.11579 and fines error from 0.37683 to 0.30110. For purity, nominal 80% interval coverage rises from 0.46667 to 0.98333 while width contracts from 0.44744 to 0.09743. Research experience produces more accurate predictions and narrower intervals that cover more reference outcomes.

A second comparison asks what predictions can be obtained from these same research histories (Fig. 5b). For the fifteen Research sessions, a source-observation mean has purity error 0.00450 and size error 0.01786, below the agent's 0.01661 and 0.07641 in all fifteen sessions. The pattern reverses for recovery: the agent's 0.11579 improves on the reference's 0.18828, with an advantage in fourteen sessions. This alternative prediction rule uses only each session's public source observations, receives no target outcomes for fitting and is scored on the same targets. A constant reference captures stable responses effectively, whereas the agent makes better recovery predictions. Research is valuable, and how its observations are used determines further response-specific advantages. The earlier corrected-crystallization cohort provides a separate supporting comparison (Extended Data Fig. 3).

The same cohort distinguishes better point prediction from better uncertainty estimates (Fig. 5c). For fines, coverage falls from 0.42778 to 0.35556 as interval width contracts from 0.68500 to 0.38450. The interval score, which penalizes width and missed outcomes, worsens from 1.4794 to 1.8047 in the world-equal mean, with worsening in every world. Research reduces average fines prediction error, but the resulting intervals express more certainty than their performance warrants. The benefit and the limitation concern the same response in the same comparison.

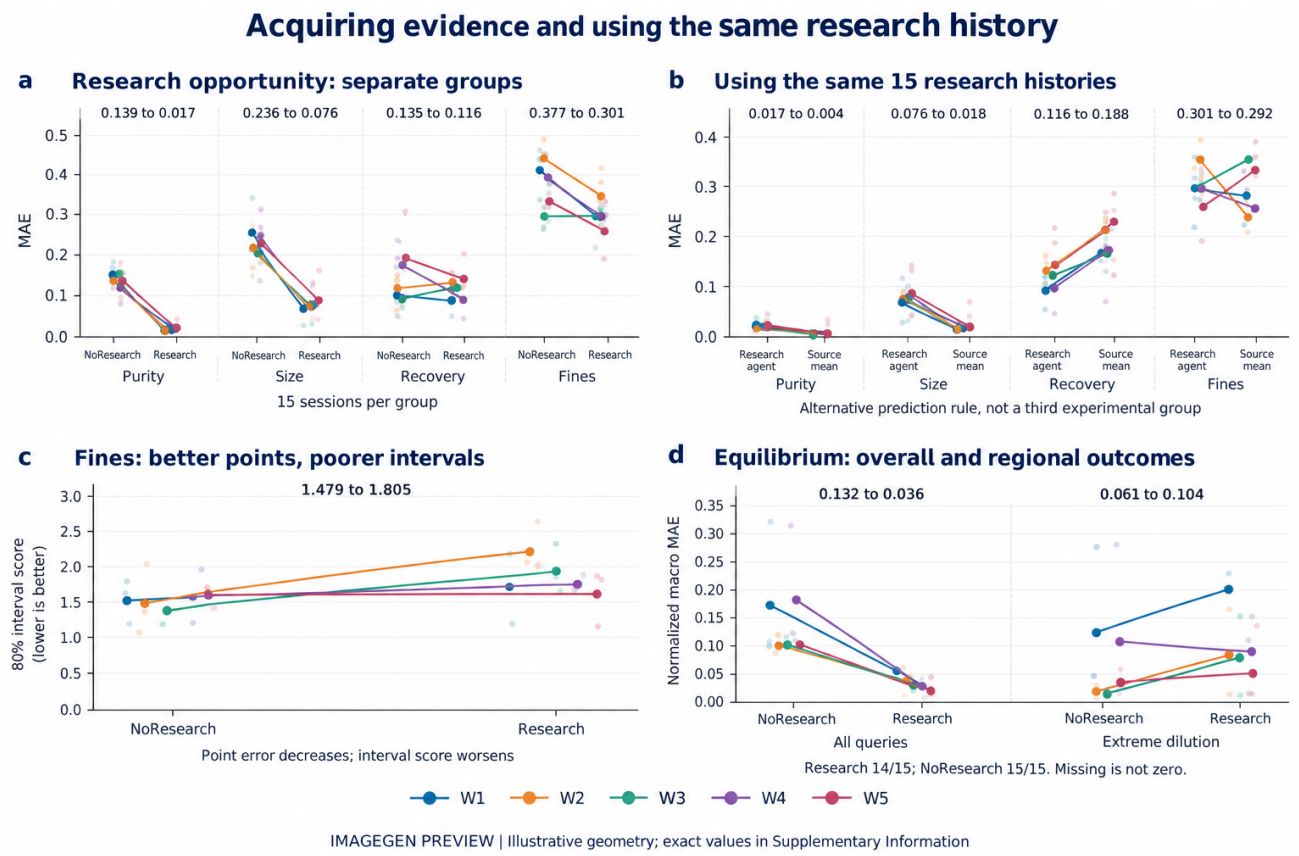



Figure 5 | Acquiring evidence, using observations and expressing uncertainty. a, Crystallization Research and NoResearch point errors, fifteen valid forecasts per arm. b, Research forecasts compared with a constant reference calculated from the same fifteen source histories and scored on the same targets; this is an alternative prediction rule, not a third experimental arm. c, Fines scores for nominal 80% prediction intervals (lower is better). d, The separate retained equilibrium comparison, overall and in the prespecified extreme-dilution region: fourteen Research and fifteen NoResearch forecasts. World three has two available Research repetitions; missing outcomes are not zero. Summary labels follow retained results. ImageGen preview; generated session dots and world-connection lines are illustrative, not exact numerical marks, within-agent learning or confidence intervals. Exact values remain in Supplementary Information.

The equilibrium opportunity comparison reveals a regional boundary (Fig. 5d). Among fourteen valid Research forecasts and fifteen NoResearch forecasts, overall normalized macro error falls from 0.132086 to 0.035632, with improvement in all five worlds. Error on the three prospectively specified extreme-dilution queries instead rises from 0.061467 to 0.103677, worsening in four worlds; the remaining nine improve from 0.155625 to 0.012950. Allowing the missing Research forecast any error in its legal range does not reverse either aggregate direction (Methods). Research improves most predictions while judgments in another region become less reliable.

Equilibrium study A makes this transition continuous in time (Extended Data Fig. 5). An initial loading-dependent expectation gives way to an observed local plateau. The researcher changes addition commands while considering alternative explanations; both commands add the same species, so the evidence establishes a local regularity without resolving the proposed chemical-identity contrast. The terminal account retains several explanations. In the later extreme-dilution forecast, it nevertheless favours a fixed acidic background and predicts dissociation 0.085, with an 80% interval of [0.035, 0.350], against reference mean 0.716032. The relationship acquired through research is useful locally; the subsequent judgment gives that experience weight in a region the experiments left unresolved.

Scientific progress has several connected dimensions: what is achieved experimentally, what the evidence establishes and what the researcher can infer under new conditions. Gains in one can coexist with unresolved questions in another. Further comparisons appear in Supplementary Information and Extended Data Fig. 1.

## Discussion

In autonomous science, agents interpret experimental results and, through their choice of experiments, shape the evidence available to them. Research goals, prior knowledge and developing judgments influence which experiments are conducted, which phenomena become visible and what can still be investigated. Continuous records in ChemWorld show researchers adapting experimental choices to observations, bringing existing judgments into new comparisons and establishing or revising scientific relationships in light of the results. Scientific understanding is therefore both an outcome of research and a basis for organizing its continuation.

These investigations reveal several substantive forms of progress in understanding. Researchers formulate

prospective expectations from observations and obtain support through new experiments. They actively test the boundaries of existing relationships and revise overly categorical judgments into more accurate empirical statements. They also establish local knowledge that guides further inference while microscopic mechanisms remain unresolved. Post-research predictions in matched chemical worlds preserve differences in their actual responses, showing that the judgments formed can reflect different properties of the worlds explored. Discovering a phenomenon, establishing a useful relationship, revising an expectation and making a supported inference thus constitute distinct scientific advances, each leaving different questions open.

Forming useful knowledge does not ensure its reliable application in every subsequent judgment. Controlled comparisons show that research can substantially improve predictive accuracy, but its benefits depend on the response and conditions examined. In crystallization, the same experimental history supports recovery predictions that outperform a simple observation mean, yet predictions of more stable responses fall short of that reference. In equilibrium, an overall predictive gain coexists with worse judgment in a particular region; better point predictions also need not bring more reliable uncertainty estimates[10,24,25]. Establishing useful relationships, judging their scope and continuing to ground subsequent judgments in evidence can therefore progress to different degrees, or even in different directions. An investigation can establish reliable local knowledge while departing from its evidential support in later generalization and application.

Controlled comparisons and continuous records together connect knowledge formation to the conditions of research. The comparisons examine the consequences of research goals, supplied information or access to experimentation; the records trace when judgments are proposed, which observations lead to revisions and how those judgments enter subsequent actions. Matched worlds allow the information actually acquired to be considered alongside final predictions, testing whether scientific judgments reflect differences in world laws. The repeated matrices concern one model configuration in fixed programmable worlds, and the process analysis draws on public judgments and executed operations. Extending this work to other models and physical laboratories, and intervening on candidate explanations during ongoing inquiry, would test the generality of these processes and the influence of particular judgments on subsequent research choices.

This study reveals concrete processes through which knowledge generation and the organization of exploration shape one another, while showing that genuine local discoveries can coexist with limitations in their subsequent application. Scientific understanding develops through experiments and influences which questions are pursued and which evidence is obtained next. ChemWorld makes this continuous process traceable and comparable, advancing our understanding of machine scientific capabilities to encompass how discoveries emerge, are revised and inform further inquiry. The central challenge for autonomous science thus extends from independently conducting research to developing reliable scientific understanding through that research and using it to expand exploration.

## Funding

This work was supported by the Shenzhen Science and Technology Program (AI2026004), the Tsinghua University Initiative Scientific Research Program and the Scientific Research Innovation Capability Support Project for Young Faculty (ZYGXQNJSKYCXNLZCXM-E7).

## References


1. G. Tom et al. Self-Driving Laboratories for Chemistry and Materials Science. *Chemical Reviews* **124** 9633–9732 (2024). https://doi.org/10.1021/acs.chemrev.4c00055.
2. D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes. Autonomous chemical research with large language models. *Nature* **624** 570–578 (2023). https://doi.org/10.1038/s41586-023-06792-0.
3. R. D. King et al. The Automation of Science. *Science* **324** 85–89 (2009). https://doi.org/10.1126/science.1165620.
4. B. Burger et al. A mobile robotic chemist. *Nature* **583** 237–241 (2020). https://doi.org/10.1038/s41586-020-2442-2.
5. B. P. MacLeod et al. Self-driving laboratory for accelerated discovery of thin-film materials. *Science Advances* **6** eaaz8867 (2020). https://doi.org/10.1126/sciadv.aaz8867.
6. N. J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature* **624** 86–91 (2023). https://doi.org/10.1038/s41586-023-06734-w.
7. A. M. Bran et al. Augmenting large language models with chemistry tools. *Nature Machine Intelligence* **6** 525–535 (2024). https://doi.org/10.1038/s42256-024-00832-8.
8. T. Dai et al. Autonomous mobile robots for exploratory synthetic chemistry. *Nature* **635** 890–897 (2024). https://doi.org/10.1038/s41586-024-08173-7.
9. A. N. Elmachtoub and P. Grigas. Smart Predict, then Optimize. *Management Science* **68** 9–26 (2022). https://doi.org/10.1287/mnsc.2020.3922.
10. G. Shmueli. To Explain or to Predict? *Statistical Science* **25** 289–310 (2010). https://doi.org/10.1214/10-sts330.
11. A. Einstein. Zur elektrodynamik bewegter körper. *Annalen der Physik* **322** 891–921 (1905). https://doi.org/10.1002/andp.19053221004.
12. I. Douven. Peirce on abduction. *The stanford encyclopedia of philosophy: abduction* (2025). `https://plato.stanford.edu/entries/abduction/peirce.html`.
13. J. R. Platt. Strong Inference. *Science* **146** 347–353 (1964). https://doi.org/10.1126/science.146.3642.347.
14. D. V. Lindley. On a Measure of the Information Provided by an Experiment. *The Annals of Mathematical Statistics* **27** 986–1005 (1956). https://doi.org/10.1214/aoms/1177728069.
15. K. Chaloner and I. Verdinelli. Bayesian Experimental Design: A Review. *Statistical Science* **10** 273–304 (1995). https://doi.org/10.1214/ss/1177009939.
16. A. Mirza et al. A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists. *Nature Chemistry* **17** 1027–1034 (2025). https://doi.org/10.1038/s41557-025-01815-x.
17. B. P. Majumder et al. DiscoveryBench: Towards Data-Driven Discovery with Large Language Models. *International Conference on Learning Representations* **2025** 4556–4579 (2025). `https://proceedings.iclr.cc/paper_files/paper/2025/file/0d70af566e69f1dfb687791ecf955e28-Paper-Conference.pdf`.
18. R. Wang, P. Jansen, M.-A. Côté, and P. Ammanabrolu. ScienceWorld: Is your Agent Smarter than a 5th Grader? *Proceedings of the 2022 conference on empirical methods in natural language processing* 11279–11298 (2022). https://doi.org/10.18653/v1/2022.emnlp-main.775.
19. P. Jansen et al. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. *Advances in neural information processing systems* **37** (2024). https://doi.org/10.52202/079017-0324.
20. K. Gandhi et al. BoxingGym: Benchmarking progress in automated experimental design and model discovery. (2025). `https://arxiv.org/abs/2501.01540`.
21. Y. Chen, P. Piękos, M. Ostaszewski, F. Laakom, and J. Schmidhuber. PhysGym: Benchmarking LLMs in Interactive Physics Discovery with Controlled Priors. *Advances in Neural Information Processing Systems* **38** (2025). https://doi.org/10.52202/085713-1556.
22. T. Han et al. MolQuest: A Benchmark for Agentic Evaluation of Abductive Reasoning in Chemical Structure Elucidation. *arXiv* (2026). https://doi.org/10.48550/arXiv.2603.25253.
23. W. Li, T. Zheng, Y. Song, G. Y. Wong, and S. See. ULTRADISCOVERY: Abductive Exploration in an Interconnected, Epistemically Open Universe. *arXiv* (2026). https://doi.org/10.48550/arXiv.2610.03092.
24. T. Gneiting and A. E. Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association* **102** 359–378 (2007). https://doi.org/10.1198/016214506000001437.
25. Y. Ovadia et al. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems* **32** (2019). `https://proceedings.neurips.cc/paper_files/paper/2019/file/8558cb408c1d76621371888657d2eb1d-Paper.pdf`.

# Methods

## Worlds, interfaces and experimental units

ChemWorld compiles compatible process components into persistent virtual chemical environments. The public contract specifies materials, legal operations, instruments, resources and task objectives. Private parameters determine each world's responses. Agents access the laboratory and public records, without evaluator state, simulator code, private parameters or withheld reference outcomes. Operations change samples; measurements may consume material. Candidate transitions are validated before commitment. Rejected operations do not constitute the requested physical intervention, although attempts can consume contract-defined allowances. Exact replay checks environment transitions, not regeneration of model decisions.

The broad study covers electrochemistry, partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. These are controlled virtual systems, not experimentally validated predictions for named physical substances. The integrated repeated design covers all six systems through complementary matrices. Five fixed worlds recur across arms and repetitions; their parameters, initial states and keyed observation mappings are fixed. Repetitions are fresh model sessions, not independent world draws or controlled model-seed replications. The model interface does not expose a usable sampling seed. Repeated identical interventions can share keyed observations, so they are not independent noise samples.

## Integrated matrices and new-session coverage

The integrated coverage design contains 450 planned researcher sessions in four matrices, combining a new 300-session block with 150 retained sessions. The objective–information matrix uses electrochemistry and reaction processing, each with five worlds × three information conditions × two commissions × three sessions, or ninety per system. The process matrix uses partitioning under discovery and purification under optimization, each five worlds × three information conditions × three sessions, plus the retained forty-five equilibrium and forty-five corrected-crystallization sessions. The opportunity matrix comprises thirty new crystallization sessions and thirty retained equilibrium sessions: five worlds × Research/NoResearch × three sessions, all Opaque. Thirty retained matched-world sessions form the fourth matrix. The retained 150 and the historical broad 300 are distinct from the new 300; a shared coverage structure does not make their protocols or periods identical.

New participants use gpt-5.6-sol at medium requested reasoning effort. Source studies allow twelve batches; the fifteen no-research participants receive none. Scientific accounts, twelve-query predictions and retrospectives are sealed in order without reference-truth feedback. The 300 planned sessions yield 288 complete chains, 298 valid forecast stages, 3,379/3,420 source final assays and 898/900 valid sealed stages. Electrochemistry completes 90/90 chains and 1,080 assays; reaction processing completes 88/90 chains and 1,080 assays, with 88 valid forecasts; partitioning completes 45/45 chains and 540 assays; purification completes 35/45 chains and 499/540 assays but retains all forty-five forecasts; crystallization completes 30/30 chains, including all thirty forecasts and 180 research assays. The new source and recommendation records have 285 and 240 verified replay receipts, respectively.

Ten purification sessions retain material-exhaustion shortfalls rather than replacement histories. One reaction forecast is absent after tool-budget exhaustion and another is invalid because its intervals violate the assessment requirements. All corresponding operations and later available accounts are retained. Four explicit service interruptions are retried after the primary queue, under prespecified technical-retry criteria; every interrupted attempt remains in the ledger and does not add a scientific unit. Earlier runtime-development attempts are separate from this block. No completed scientific outcome is replaced because of its score.

## New-matrix measurements and aggregation

Each response retains its native assessment definition. Electrochemistry and purification use a single keyed final observation as reference; partitioning uses noiseless presampling phase fractions; reaction processing uses a five-observation mean for point error and all sixty reference observations across twelve queries for coverage; crystallization uses noiseless preassay responses and initial-charge recovery. Partitioning intervals are nominal 90%, whereas the other systems use 80%. The two partition fractions are complementary, not independent evidence. No cross-response or cross-system scientific-ability score is formed.

Response errors and interval diagnostics are computed per session. Within a condition, available sessions are averaged within each fixed world and the five worlds receive equal weight. The electrochemical commission comparison further averages the three information conditions within each world; all forty-five forecasts per commission are available. The reaction Opaque–Aligned optimization comparison uses fifteen valid forecasts per condition. Other reaction cells with missing forecasts retain fourteen of fifteen and are explicitly marked. Purification uses available forecasts even when the source budget is incomplete. Full-grid estimates remain unavailable when planned responses are missing; available-case values do not impute zero error. Repeated sessions, queries and responses share worlds and are not treated as independent

draws for confidence intervals or significance tests. Three repetitions per fixed cell describe variability, not a population reliability rate.

Operating comparisons use independent retests of sealed recommendations. They are distinct from predictions under new query interventions. Scores and yields retain their public definitions and are averaged separately. Interval width, empirical coverage and, where present in the retained evaluator, the proper interval score are reported alongside point error. For the new crystallization opportunity comparison, the score uses alpha = 0.2 and its legal response scale. Research and NoResearch have the same scientific framing and sealed assessment, with research additionally supplying an experimental budget, experience and recording opportunity. Their difference estimates this complete package, not a specific reasoning operation.

### Full reading of the new research histories

All 300 new sessions are read, including the fifteen no-research accounts, all 873 notebook versions and 31,922 public operations. The evidence contains 899 nonempty sealed payloads: one includes an invalid forecast, so there are only 898 valid stages; one forecast payload is absent. Coverage comprises electrochemistry (268 notes; 7,647 operations), reaction processing (293; 7,917), partitioning and purification (262; 13,253), and crystallization (50; 3,105). The retained per-session readings verify 1,363 quotations against public records. Missing notes, unsuccessful tests, rejected requests, terminal revisions and later retrospective admissions remain visible. Reading coverage is not an explanation-success denominator.

For each focal judgment, analysis distinguishes the stated phenomenon, candidate explanation or empirical expectation, its first recorded appearance, prospective consequence, actual committed intervention, measurement, revised standing and later use. A conditional plan is not counted as an executed test. Subsequent predictions are not retroactively inserted into earlier notes. A terminal explanation can support a claim about later prediction, but cannot by itself establish an earlier influence on source actions. Source-time public contracts and the frozen execution semantics distinguish rejected requests, rolled-back transitions and task-quality failures. In particular, a reaction-risk threshold does not imply rollback, and direct heating and controller configuration have different public temperature limits. Quench changes temperature and risk and advances the clock without additional reaction integration; waiting before quench may continue chemistry. Reaction study A compares different cooling histories, so its quench-before-wait result supports a local operational relation without separating thermal and nonthermal effects. The selected crystallization seed review compares requested and committed amounts, rather than treating repeated rejected requests as dose.

All histories receive semantic reading, while illustrative claims are selected retrospectively for their explanatory relevance. Multiple claims can coexist within a session with different support and outcomes. We trace individual judgments within complete trajectories, retaining session and world levels for comparisons. The analysis does not infer private cognition or assign one success label to a whole session. Interpretation and source verification are performed within the assistant analysis workflow; there is no independent human coding, blinding or second-rater reliability estimate. The earlier action-linked windows and selected explanation histories below retain their original definitions and denominators.

### Same-cohort source references and execution-state interpretation

We additionally summarize the source-only references already stored with the fifteen new crystallization Research results. This is a post hoc analysis of retained outputs, with no model or simulator execution. The constant reference averages each session's response-complete public final-assay records; the existing nearest-neighbour reference selects the source recipe nearest to the query in the original public recipe-feature space. Targets are used only for scoring. Response completeness requires numeric values for all four assessed responses, irrespective of their values or task success: 179 of 180 source assays meet this rule; one assay in world 1, repetition 3 lacks the full response vector, leaving eleven fitting assays there. All fifteen sessions and 180 queries per response remain in the analysis. We report mean absolute error against the same noiseless preassay targets as the agent, averaging sessions within worlds and then worlds equally. These point-only references assess use of available observations and are distinct from the NoResearch arm. Per-session errors, fitting counts and source means are retained in the portable archive. This is a comparison of prediction rules on identical histories, not a third experimental group, a measure of private understanding or a causal decomposition of the research-opportunity effect.

For the supporting heating-reconstruction case, the visible return contains outer `ok: true` and `experiment_completed: true`, but also `transaction_status: rolled_back`, an explicit rollback error and zero changes in temperature and time. The two requested 600 s operations therefore do not establish 1,200 s of chemical exposure. The conflicting public labels are preserved alongside the later account and forecasts; this case does not isolate agent reasoning from interface ambiguity. Another session explicitly excludes rolled-back heat from chemical interpretation. Controller and direct-heating limits are checked against their distinct public contracts. These execution checks locate evidential differences, rather than treating every discrepancy as a chemical misconception.

### Historical and retained evidence layers

The historical broad pool contains 240 Sol-medium campaigns and 60 additional equilibrium campaigns from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort. The five-model equilibrium comparison includes fifteen Sol campaigns already in the 240. Requested model labels do not establish equal realized computation. The original 240 include 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium, 30 crystallization and 15 purification campaigns. They retain 3,597/3,600 final assays, all 720 assessment stages and 238 protocol-conforming campaigns under their original rules. The additional sixty retain 720 batches and 240 stages. Two original source-completion failures remain included. Earlier selected notebook studies comprise another 24 sessions and are not pooled into the broad or retained equilibrium–crystallization process cohort.

The retained equilibrium–crystallization process cohort combines the 45 equilibrium sessions from the earlier 48-session block and its 42-session extension with a complete replacement block of 45 crystallization sessions under a consistent recovery description. Each system covers five worlds, Opaque/Aligned/MisIndexed information and three sessions per cell, with twelve research batches per session. Equilibrium retains 39 complete chains and 41 valid forecasts; corrected crystallization retains 41 and 42, respectively. The combined retained denominator is 90 sessions, 80 complete chains and 83 valid forecasts. Old crystallization sessions remain protocol-affected historical evidence. They are not added to the replacement sessions when calculating current outcomes or process frequencies.

Corrected crystallization has 510/540 source batches, 125/135 assessment stages, 145 notebook versions and 42 recommendation retests. All 45 source trajectories and 42 retests pass replay. Four incomplete chains remain: three lack forecasts, and one retains a valid forecast but lacks the subsequent retrospective. The equilibrium subset and the separate opportunity block retain their own failure tables. Missing outcomes are not set to zero, and later favourable outcomes do not replace completed results. Evidence integration does not change the original development or qualification status of any block.

### Information, autonomy and recording

Historical goal comparisons pair independent discovery and optimization campaigns within world, information arm, batch budget and prior locus. Electrochemistry contributes thirty pairs across twelve- and twenty-four-batch conditions; reaction processing contributes thirty across parameter- and structural-prior blocks at twelve batches. Operating outcomes are independent retests of sealed recommendations. The joint-outcome counts use public-score prediction MAE, whereas the reaction prior comparison uses the mean across its six normalized response errors. These outcomes are not interchanged. Budget comparisons likewise concern independent sessions with complete resource envelopes; they do not estimate the effect of continuing the same investigation.

The historical five-model equilibrium comparison contains five worlds, three information arms and one twelve-batch campaign per cell: seventy-five campaigns, including the fifteen Sol campaigns already counted in the broad study. The 1 mM threshold descriptively summarizes the lowest positively loaded nominal concentration assayed; it is not a discovery criterion. Concentration is based on nominal recipe loading, and the three dilute assessment queries are shared across configurations. Forecast errors, original prediction intervals and acquired evidence are retained for all arms. The selected Astra and electrochemical trajectories are retrospective process illustrations, with scientific accounts written after experimentation; they are not contemporaneous notebook records or additional replications. Complete historical design, response tables and case values remain in Supplementary Information.

Opaque supplies the public task and operational context without the additional instance dossier. Aligned supplies partial information appropriate to the system. In equilibrium, this includes an effective-acidity interval and an anchored local dilution relationship. MisIndexed uses another world's archival relationship with the same format and qualification. In crystallization, Aligned gives nominal descriptors for anonymous materials; MisIndexed swaps solvent S1/S3 descriptors while preserving S0, S2 and catalyst descriptors. Neither dossier specifies the complete world law.

The original equilibrium Opaque/Aligned sessions precede its MisIndexed extension. Corrected crystallization is a later complete block. These descriptive comparisons are not all concurrently randomized treatments, and new versus old crystallization is not an estimate of the causal effect of correcting the prompt. The retained equilibrium opportunity comparison uses a common contemporaneous protocol, described below.

Agents choose operations, quantities, measurements, repetitions and grouping within fixed budgets. Equilibrium permits twelve source batches, 180 operation attempts, twelve intermediate instruments and twelve final assays; crystallization uses its fixed twelve-batch materials envelope and 720-attempt ceiling. A notebook role asks for an initial testable expectation when available; subsequent updates are agent-selected, without a write-every-batch requirement or automatic reinjection. The role may influence research and is not assumed passive. Notes retain version and public-event cursor. These public records do not expose private cognition. Absence of a notebook-read tool call does not establish absence of access to the conversation's earlier content.

### Assessments and recovery definition

After source research and any operating recommendation, agents seal a scientific account, answer twelve fixed new-intervention queries without laboratory tools and then reflect without truth feedback. Equilibrium includes a structured supplemental assessment. The evidence notation K1/Q/K2/EQS denotes these stages. Queries are hidden during research; post-assessment proposals are not treated as executed source experiments.

Equilibrium errors compare predictions with the mean of five reference observations per query. Coverage checks all five observations. Crystallization uses the noiseless full-process response immediately before final-assay sampling; source observations are noisy. The corrected block retains sixty reference batches, numerically matching the earlier reference outcomes, and separately retests each available committed recommendation.

The corrected recovery definition is seed-excluded recovered solid target divided by the tracked initial reagent charge, in moles, clipped to [0,1]. Source instructions, prediction prompts, observations and scores now refer to this same full-process quantity. The internal crystallization-stage metric instead divides by target product before separation. Earlier process instructions described this latter denominator while observations and scoring used initial reagent charge. Original scores are preserved as protocol-affected diagnostics; they do not enter the corrected cohort. That corrected block was rerun in full rather than retrospectively relabelling those scores.

Quality requires particles, purity at least 0.80 and fines at most 0.50; joint recommendation success also requires corrected recovery at least 0.10. Corrected joint counts are 13/15 Opaque, 10/14 Aligned and 10/13 MisIndexed among available retests, against fifteen planned sessions in each arm. Particle size is the public index min(d50/250 micrometres, 1); its error is not measured in micrometres.

### Research paths and process interpretation

We trace publicly stated relationships and expectations, actual experiments, acquired observations, revisions and subsequent uses. A scientific account or an accurate forecast alone cannot establish this sequence. Interpretation is based on specific histories, not a general reasoning-ability scale. Five named core histories come from the new block: Purification studies A and B, Reaction study A and Electrochemical studies A and B. Supplementary Information maps their stable labels to information conditions, complete public records, note revisions, operation positions and batch sequence. Matched-world studies A and B retain their separate block's identities and anchors. Selection follows complete reading and remains retrospective. The earlier 92 action-linked windows describe recording coverage in the retained equilibrium–crystallization cohort; the equilibrium opportunity case is outside that window cohort.

For the retrospective explanation-history panel, selection is fixed to the focal equilibrium opportunity session and four cases from the corrected-crystallization cohort: lower charge, interpolation, and the two contrasting seed accounts. We trace seven related judgments across these five sessions, retaining all notebook versions, including terminal revisions. Fields distinguish the phenomenon, a candidate or expectation actually stated by the agent, first appearance, conditional prediction, executed operations, observations, subsequent standing and later use. Missing conditional statements are left missing rather than supplied by the analyst. The equilibrium loading expectation, component hypothesis and background interpretation are connected strands of one inquiry, not three independent discoveries. The panel is not sampled to estimate frequencies; no explanation-level prevalence or reliability denominator is inferred from the window counts.

The focal operation-semantics check uses the retained execution version and trajectory's mechanism identity, not current runtime assumptions. That world's single non-catalytic feed is selected both by the reagent operation and by component 0; both add to the same initial-amount ledger read by the equilibrium response calculation. Switching these commands at equal loading therefore does not switch chemical identity. Final nominal concentrations in Extended Data Figure 5 divide committed added moles by total added solvent volume; the staged-dilution batch is plotted at its final 0.8 mol/L. No new simulator calls are required for this inspection. Proposed low-loading and orthogonal measurements remain unexecuted in the retained source history. The scope of the resulting plateau is assessed independently of the agent's proposed mechanism.

Retrospective path extraction covers sixty historical electrochemical, forty-five equilibrium and forty-five corrected crystallization sessions. Fifty within-world condition contrasts retain goal, budget, information and cohort. Counts use committed operations; solvent coverage includes additions during resuspension. Rejected attempts do not establish successful material use. Interrupted sessions remain in the planned denominator.

Structured extraction considers all 178 adjacent pairs among 268 notebook versions in the retained 90 sessions. Eligibility requires an earlier note, at least one new committed final assay before the next note, and a committed research operation after that later note; measurement and termination alone do not satisfy continuation. All 178 pairs contain new assays. Ninety-two have the required continuation: 34 equilibrium and 58 crystallization windows in 30 and 35 sessions, respectively. The other 86 pairs (44 equilibrium, 42 crystallization) have no later committed research action and remain available in the evidence archive. The 25 sessions without a complete window are not classified as having learned nothing.

Eligibility is computed from event timing before semantic plan matching, without testing whether a proposed action succeeds. Nevertheless, the rule conditions the sample on continued committed research and excludes terminal revisions from this particular action-linked denominator. All 92 eligible windows are coded. Labels concern one selected immediate plan,

not every proposal in a note: 83 matched, seven partly matched and two without an explicit next plan; corrected crystallization contributes 52 matched and six partly matched. Full notes retain abandoned, rejected and altered proposals even when another selected action matches. All 182 update/plan quotations and their assay/action cursors are verified against source records. The archive includes every adjacent pair, selected code, full note and execution history.

A further internal review covers the five named new histories and two named matched-world histories. It checks exact quoted wording, note cursors relative to executed comparisons, operation arguments, measured outcomes and later accounts or forecasts. The public packet retains the full notes, batch actions and scientific posttests alongside a supported finding and scope boundary for each case. This review uses the same assistant workflow, with prior knowledge of the outcomes; it is not blinded or independent.

Internal review is retrospective and unblinded, without an independent second-rater agreement estimate. Counterexamples include warranted retractions, correct recognition of unexecuted tests and accurate but explicitly conjectural extrapolation.

### Matched chemical worlds and explanatory histories

The additional prospective block contains five parameter pairs, two chemical networks and three fresh Sol-medium sessions per world: thirty planned sessions. Network A couples weak-acid dissociation to direct free-ion precipitation; B adds neutral aqueous ion pairing. Within each pair, acid dissociation, solubility-product, cation, activity and initial-state parameters are matched, along with the public operational contract. Two actual starter assays, each using $6 \times 10^{-7}$ mol in 0.080 or 0.075 L, are supplied before autonomous research. The paired noise-free responses differ by no more than one declared final-assay standard deviation in each channel. Participants receive chemical framing and loading, dilution and scale dimensions, but not the private network identity or parameters. The initial notebook therefore follows supplied observations, not an evidence-free pretest.

Qualification of the implemented candidate covers 230 physical states and assays, thirty exact replays and 120 runtime mean checks. Five bounded nuisance refits do not adequately fit network B with the tested direct-network family; this check does not prove identifiability against all alternative explanations. An earlier failed physical candidate remains separate. These checks concern the world's implemented information structure, not agent learning.

Each session has twelve autonomous batches followed by a sealed scientific account, twelve query forecasts and a retrospective without truth feedback. The queries are withheld during research, although agents can visit corresponding physical conditions. All thirty sessions complete all stages: 360 source final assays, ninety sealed stages and 1,080 query-response predictions. Eighty notebook versions comprise twenty sessions with three notes and ten with two. Thus continuous midpoint revision cannot be assumed in every session. Three fresh sessions per world share fixed observation keys; the model sampling seed is unavailable.

Every committed source action sequence is replayed in its own and the paired world: thirty source replays and sixty own/partner trajectories pass. Separation is the absolute difference between noise-free response means divided by the final-assay standard deviation: 0.002 for normalized pH and 0.006 for each fraction. Of 360 final assays, 326 separate at least one channel by three standard deviations and 158 separate at least two; all thirty sessions include an assay of the latter kind. This threshold describes physical information, not statistical significance or the researcher's recognition of a mechanism. Counterfactual replay fixes actions and does not estimate how an agent would adapt to different observations.

Forecast errors use the mean of five retained reference observations per query; intervals are evaluated against all five observations, which are shared across sessions in the same world. Session values are averaged within each world, then the ten worlds equally. Intervals are nominal 80% prediction intervals, not confidence intervals for means. Their high empirical coverage must be read alongside their widths and does not by itself establish calibration. This block has no no-research arm and uses different chemical laws from the opportunity comparison.

A post hoc forecast-contrast analysis retains all 1,080 predictions. For each of five parameter pairs, twelve common queries and three responses, it subtracts the mean of three A-world forecasts from the mean of three B-world forecasts and compares that difference with the difference between five-observation reference means. All 600 executed reference action sequences match the common query definitions. All 180 contrasts remain, including zero, tied and opposing differences. Magnitude accuracy is the mean absolute error between forecast and reference differences across all sixty contrasts per response. The zero-difference reference gives identical forecasts in both worlds, so its error equals the mean absolute reference difference. Relative error reduction is one minus the ratio of these aggregate MAEs; there is no per-query division by small gaps. We report errors separately for each parameter pair to expose variation across the five pairs.

A descriptive subset uses reference gaps of at least three single-assay standard deviations (0.002 for pH and 0.006 for fractions), a physical scale rather than a significance threshold. Query contrasts share worlds and several target concentrations; each world contributes three fresh sessions with uncontrolled model sampling seeds. The analysis measures final forecast sensitivity to world differences. Supplied starter observations are close but not identical, and there is no no-research arm, so it does not identify a change from baseline or isolate the research effect.

All eighty notes and ninety sealed stages are read with their executed actions. Sixty-seven selected focal histories retain 365 verified quotations. A separate retrospective reading covers 105 existing sessions and 309 notes, tracing 112 selected

histories and 218 note anchors; four selected sealed follow-ups add thirteen anchors. The 105 are the retained ninety plus fifteen research-opportunity sessions, not additional experiments. Histories can overlap within a session and include terminal revisions. They are descriptive units selected for explanatory relevance, not independent discoveries or population frequencies. Initial expectations, candidate explanations, empirical relations, executed tests and subsequent uses remain distinguishable; absent causal or conditional statements are not supplied by the analyst.

Source operations retain 1,901 attempts: 1,893 committed, six rolled back and two rejected by validation. Another 131 recovered tool failures are counted separately. The first launch attempt contains four startup failures, twenty-six unstarted sessions and no scientific operations; after a thread configuration repair, the entire thirty-session block restarted. All earlier failures and resource records are retained. No completed scientific outcome is replaced because of its value.

### Source-only forecast references

Two transparent references are computed for each valid corrected-crystallization forecast using only that researcher's successful source final assays. The first averages all available assays. The second averages assays whose exact set of committed solvent identifiers matches the query, falling back to the overall mean when none match. Solvent sets include successful solvent additions and resuspension, excluding rejected actions. The fitting function receives no test truth. These references are descriptive calculations, not separately trained agents or mechanistically complete baselines.

There are 42 valid sessions and 504 queries. Same-solvent fallback occurs in 186/504 queries: 1/180 Opaque, 95/168 Aligned and 90/156 MisIndexed. Matching does not control thermal history, quantities or final liquid composition. References have no predictive intervals. Each response is reported in its own public scale; no combined score is formed across recovery, purity, size and fines.

The added distribution analysis uses those same 42 valid sessions and their 504 source final assays. For each response we expose source observations, all query reference means, forecasts, signed errors and session summaries. Descriptive pooled means and quantiles retain repeated query targets; they are not independent-condition estimates. Signed error is forecast minus reference mean. The world-equal signed mean first averages twelve queries within each session, then sessions within each world and finally the five worlds. Three missing forecasts remain in the planned denominator of 45. These calculations use only the corrected cohort and do not import purity levels or directional bias from historical crystallization.

### Retained equilibrium research-opportunity comparison

The fixed design contains five equilibrium worlds, Research/NoResearch and three fresh Sol-medium sessions per cell, for thirty planned sessions. All use Opaque information. The shared public background contains the task and measurement definitions, material catalogue, legal operations/instruments and initial state; the instance archival prior is null. Opaque therefore means absence of an additional instance-specific law, not absence of public chemical framing or model pretraining. The complete initial public contexts accompany the manuscript. Fifteen within-world/repeat checks confirm scientific-input equality after removing only the non-scientific campaign identifier. Research additionally receives twelve autonomous source batches and the existing notebook opportunity; NoResearch receives no source operations or observations. Both arms use identical assessment prompts, schemas, order and restrictions.

All thirty sessions reach terminal status. Research completes 180/180 batches; the combined arms complete 117/120 assessment stages and provide 29 valid forecasts. In world three's third Research repetition, the forecast stream disconnects after the scientific account. Q and the two later stages are missing, without replacement. Thus Research has fourteen valid forecasts, NoResearch fifteen; Research world three has two rather than three. Fifteen source trajectories and twenty-five reference trajectories (300 reference batches) pass exact replay. The retained resource ledger is complete for 29/30 sessions; one has only recorded partial usage. Recorded input plus output totals 13,428,656 tokens. Missing usage is not zero and monetary cost is unavailable.

A first execution attempt failed because the assessment transport configuration omitted a required field. Four launched sessions, two completed source histories totalling twenty-four batches and their resource costs are retained separately; no scientific-account stage completed. After fixing that execution defect, the whole fixed thirty-session block restarted. This was a protocol repair, not replacement based on scientific outcomes. Only the complete restarted block enters the comparison.

The comparison estimates the total research-opportunity package under a shared assessment. It does not isolate autonomy relative to supplied data, feedback alone, notebook use or private representational change. It is between fresh sessions, without a within-session pretest. World and arm order are balanced in a predetermined launch schedule, not a random sample of world laws.

### Quantitative summaries and missingness

Equilibrium macro MAE equally weights normalized pH, dissociation fraction and precipitation signal across the designated queries. The extreme-dilution group comprises three queries fixed before the repeated blocks; the remaining nine are not assumed entirely in distribution. Session errors are averaged within each world, then the five worlds receive equal

weight. Individual sessions and all missing stages remain visible. Responses, queries, reference observations and batches are correlated within sessions and worlds; they are not independent researcher samples.

Coverage and mean interval width accompany the proper interval score for the agents' nominal 80% intervals[24]. With lower endpoint l, upper endpoint u, outcome y and alpha = 0.2, the score is (u-l) plus (2/alpha)(l-y) when y is below l, or (2/alpha)(y-u) when y is above u. Lower is better. Aggregates weight worlds equally. Agent intervals are not confidence intervals for our group means.

A post hoc sensitivity analysis assigns the single missing opportunity forecast any macro MAE in [0,1], the legal error range, and recomputes means using the planned three sessions per arm/world and five equal worlds. Research-minus-NoResearch bounds are [-0.098659, -0.031992] overall and [+0.036768, +0.103434] for dilution. These are conservative bounds on missing-outcome influence, not imputation, confidence intervals or population inference. Three repetitions per fixed condition describe variation without establishing a population reliability rate.

### Matched quench intervention

A separate retained block applies the eighteen original first-cohort crystallization recommendations in five worlds, with and without a single quench operation: ninety pairs, 180 trajectories. All other action arguments and initial states are matched. Quench is removed at its recorded location or inserted immediately before the first seed/crystallization-cooling operation. No agent adapts or receives these outcomes.

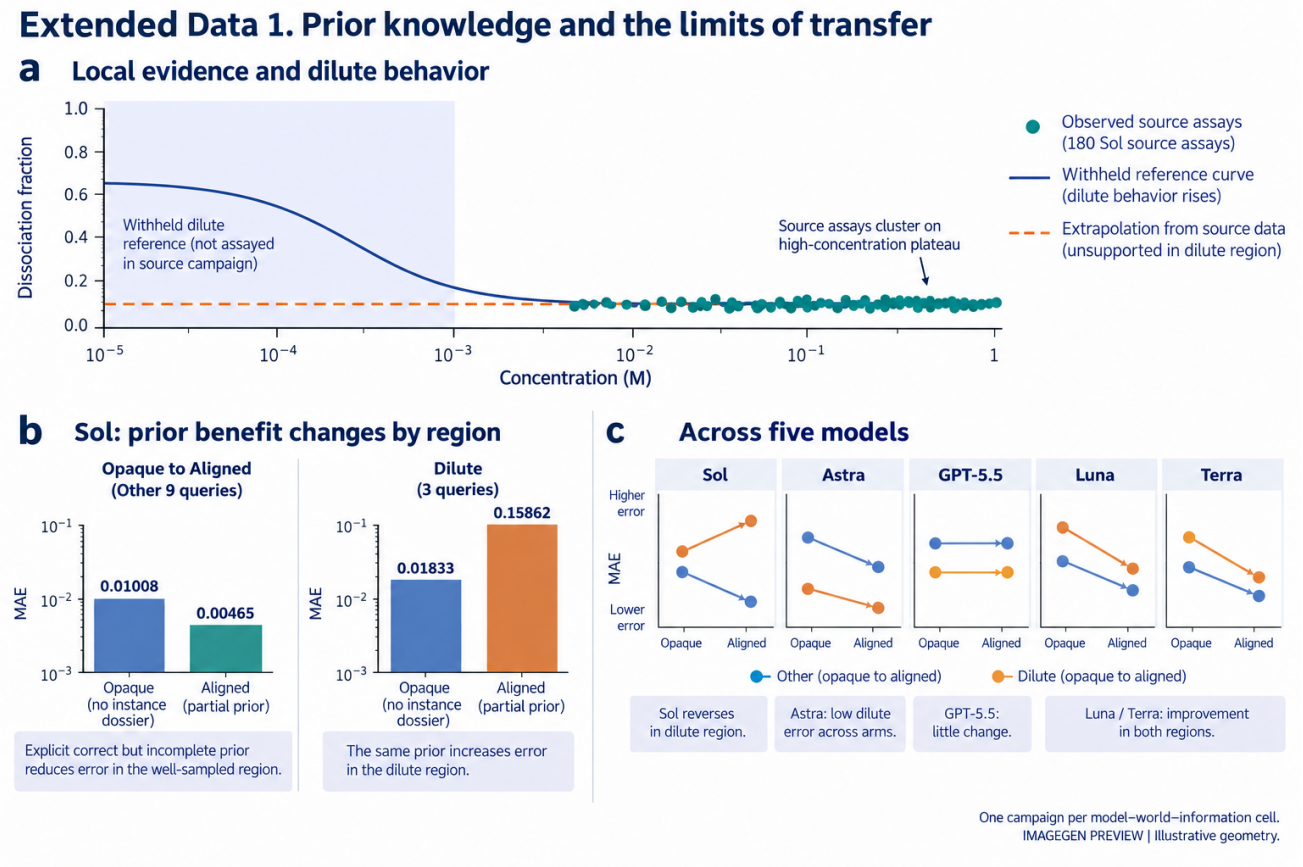
All trajectories are retained and replayed; seventy-five pairs reach both final assays. In five, the absent arm lacks particles; in ten, the present arm makes the subsequent fixed cooling request invalid. Failed arms are not repaired. Among complete pairs, mean present-minus-absent fines changes by +0.30937 and recovery by +0.00741; joint success changes from 34 to 4. These conditional effects describe physical operations under the implemented initial-charge recovery metric. Quench changes multiple physical state variables and does not isolate cooling rate or a cause of scientific judgment. Donor recipes are historical and are not sampled from the corrected block.

### Data and code availability

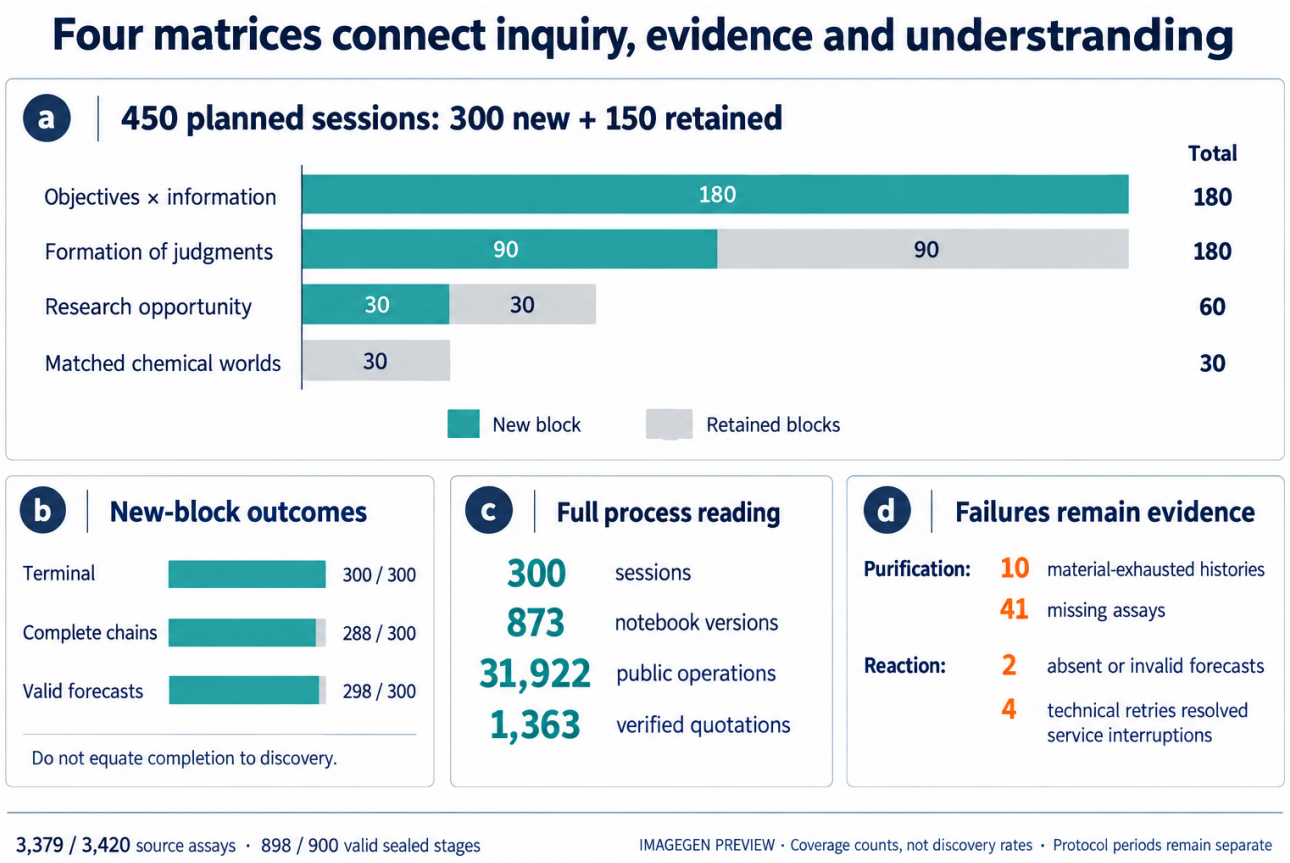
A portable review archive contains the new 300-session machine summaries, all 1,380 planned session–response rows, recommendation outcomes, failures, full curated public histories, semantic readings and quotation anchors. It includes all five system-level coverage tables and the four-matrix design. The retained equilibrium opportunity data, all 178 adjacent note pairs from the older ninety-session process cohort, corrected-crystallization distributions and thirty matched-world histories remain separately accessible. Missing responses remain explicit. Public scientific payloads are included; raw provider envelopes and credentials are excluded. Original replay and resource records remain in the project evidence locations. A persistent public deposit remains to be finalized.

All five main figures and eight Extended Data figures are labelled ImageGen previews. Numerical labels refer to retained results; generated geometry, curves and session dots are illustrative. Tables and source records supply exact values, and data-derived reference plots for Figures 2, 4 and 5 are available separately. Supplementary Information identifies each experimental block and includes all thirty-four historical numerical tables.

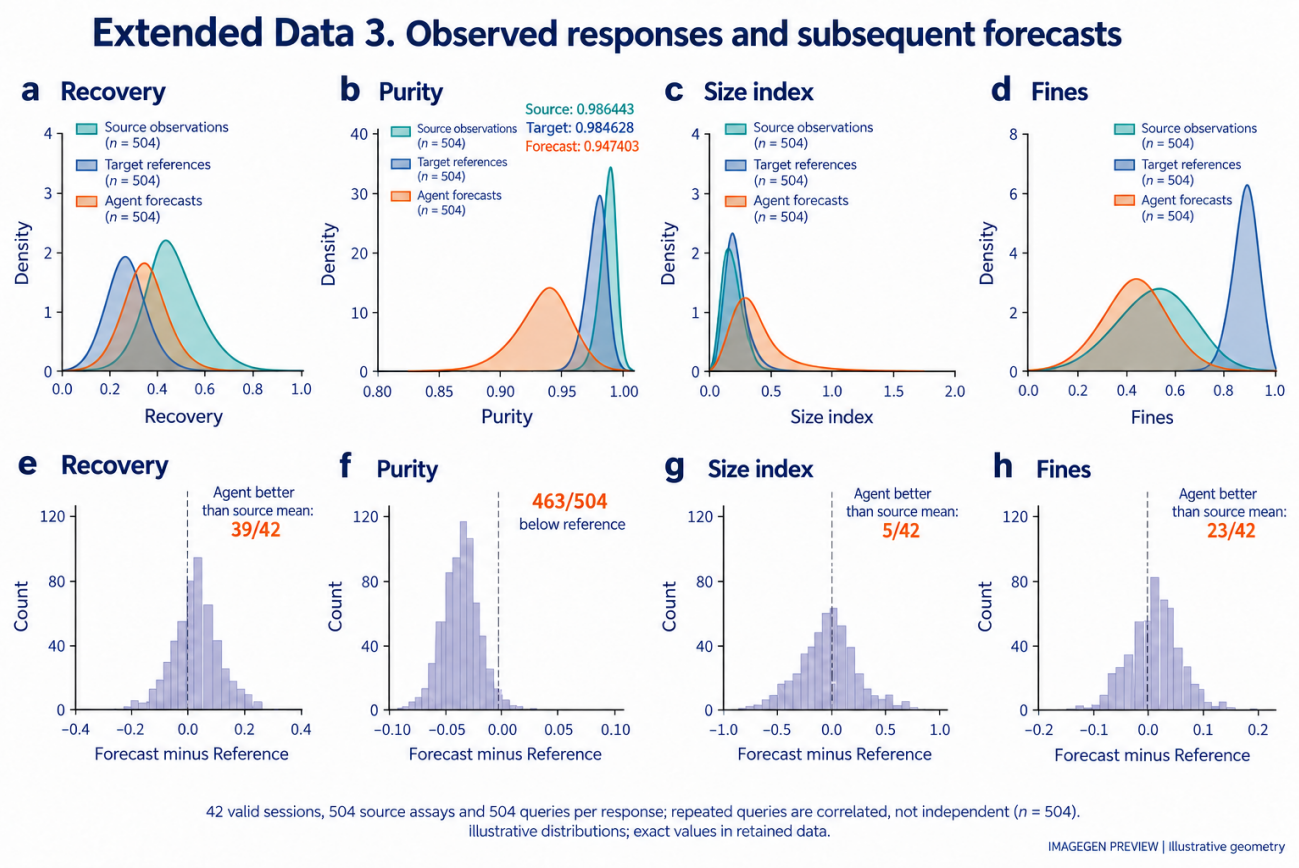
## Extended Data



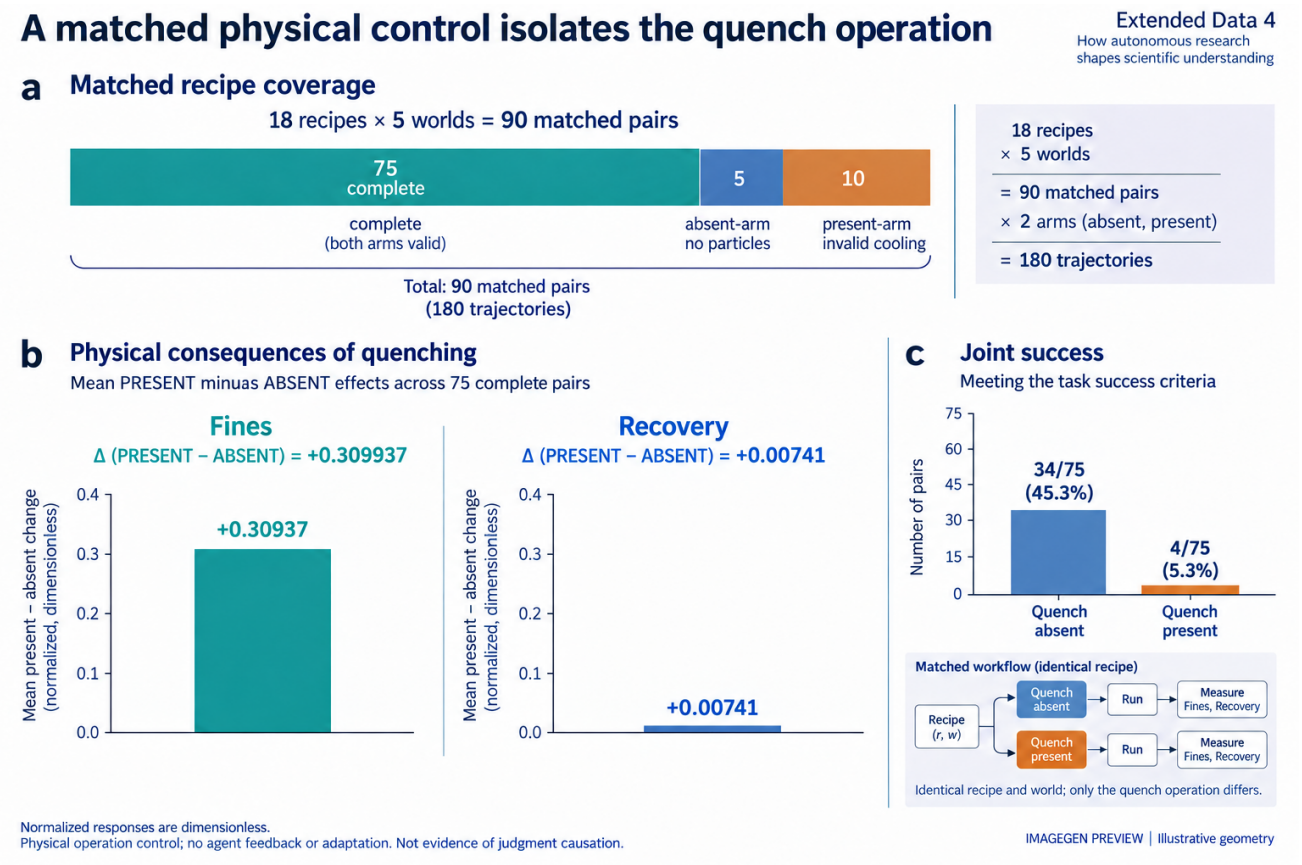

Extended Data Figure 1 | Partial prior knowledge and transfer. a, Schematic local plateau, dilute reference behavior and an unsupported continuation. b, Historical Sol/Opaque versus Aligned errors in two query regions. c, Qualitative patterns across five models, not numerical effect estimates. Historical cells contain one campaign each; the dilute partition was post hoc in that cohort and fixed before subsequent repeated blocks. Repeated-cohort estimates are not pooled here. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



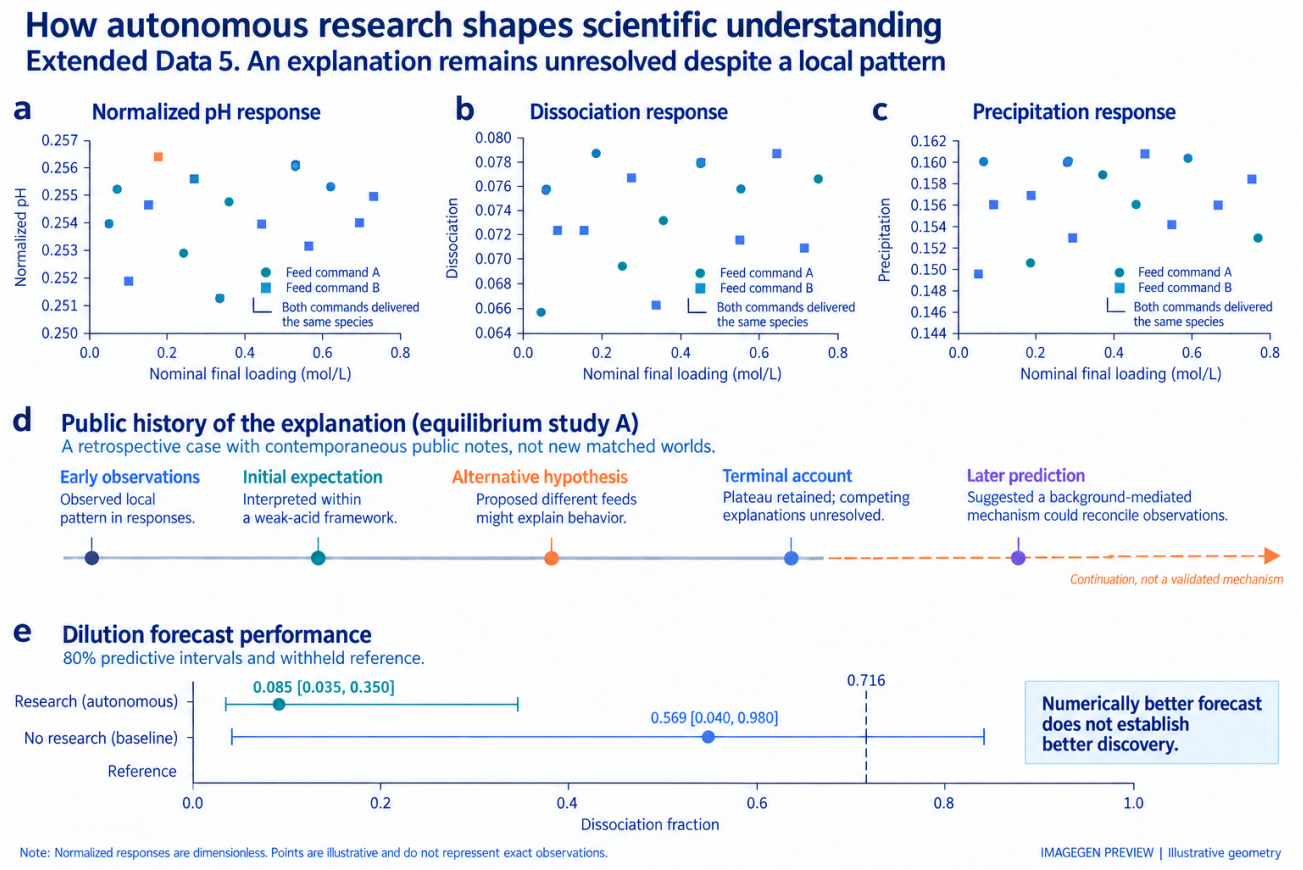

Extended Data Figure 2 | Integrated design and new-block coverage. a, Four matrices contain 450 planned sessions: 300 new and 150 retained under separate protocols. b, The new block retains 288 complete chains and 298 valid forecasts; completion is not scientific success. c, Full reading covers 873 notebook versions and 31,922 public operations, with 1,363 verified quotations. d, Ten purification histories miss forty-one assays after material exhaustion; two reaction forecasts are absent or invalid. Technical retries resolve four service interruptions without replacing scientific outcomes. ImageGen preview: geometry is illustrative.



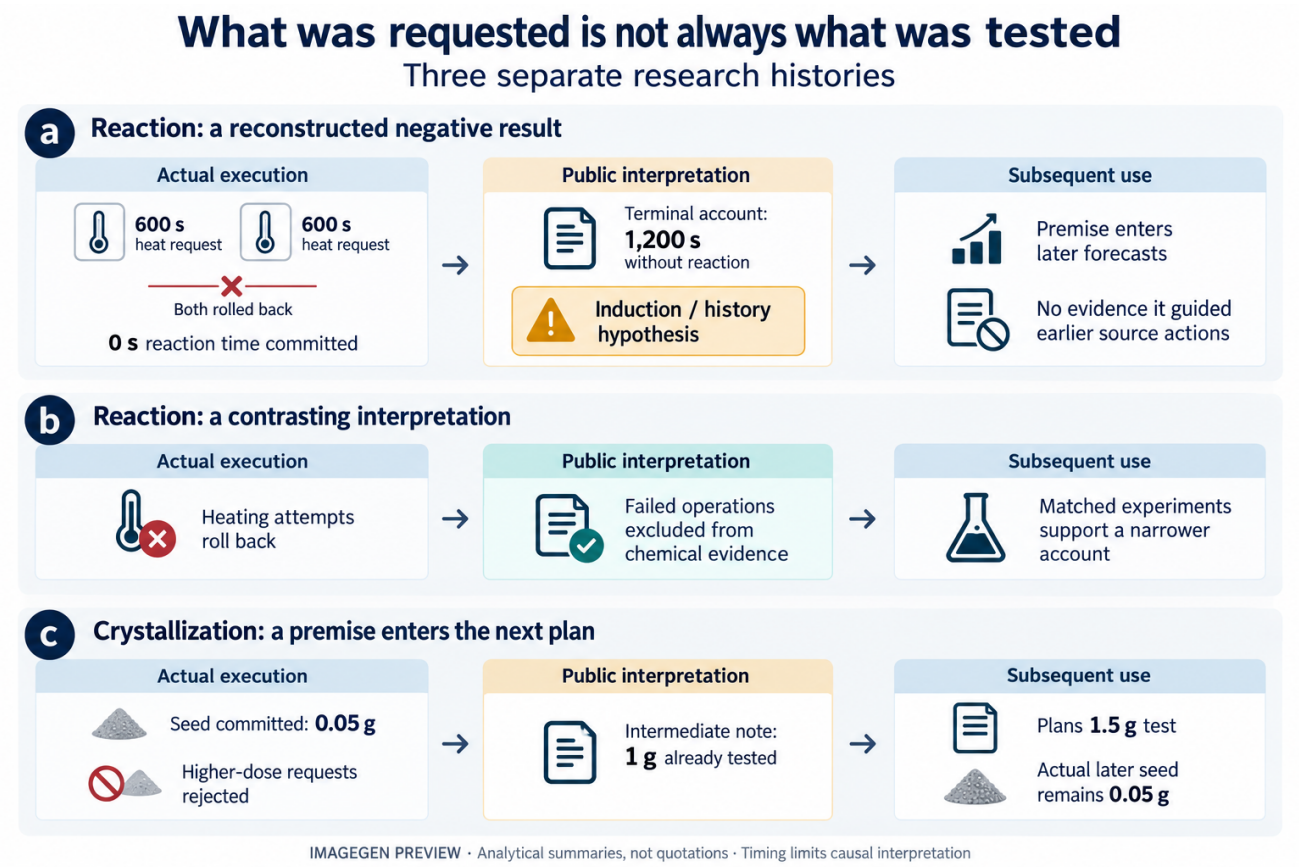

Extended Data Figure 3 | Corrected-crystallization response distributions. a–d, Illustrative distributions for source observations, reference targets and agent forecasts. e–h, Illustrative signed-error distributions; their geometry and bin counts are not observations. The retained cohort has 42 valid forecasts, 504 source assays and 504 queries per response; shared targets are not independent researcher samples. Numeric callouts retain source summaries. Size is a dimensionless index. Three missing forecasts remain in the planned denominator of 45. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



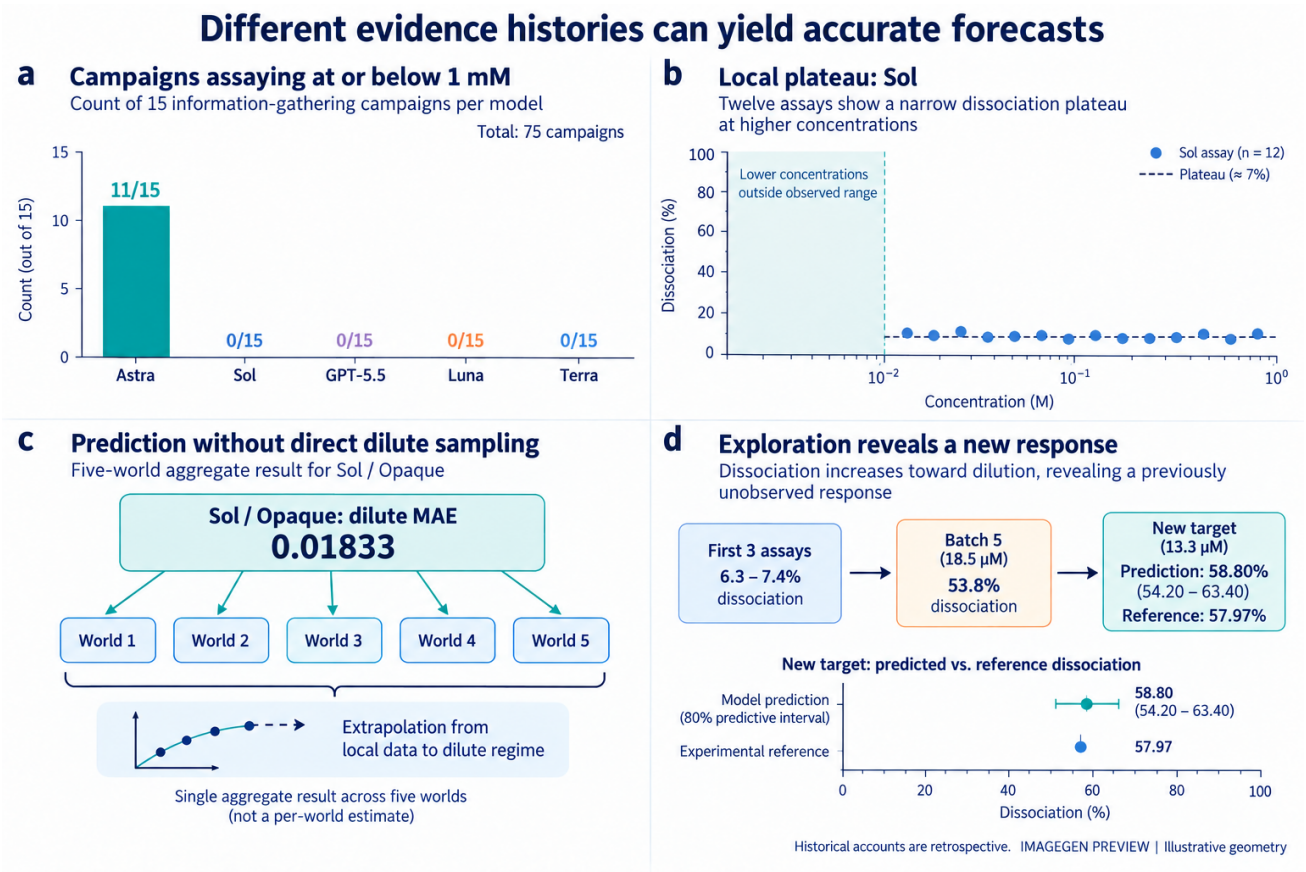

Extended Data Figure 4 | Matched physical effects of quenching. a, Eighteen historical recipes across five worlds yield ninety pairs, with seventy-five complete, five absent arms lacking particles and ten present arms invalidating cooling. b, Mean present-minus-absent changes among complete pairs. c, Joint task success in those 75 pairs. These are physical effects under the implemented initial-charge recovery metric. Agents neither adapt nor receive these results, so no effect on judgment formation is inferred. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



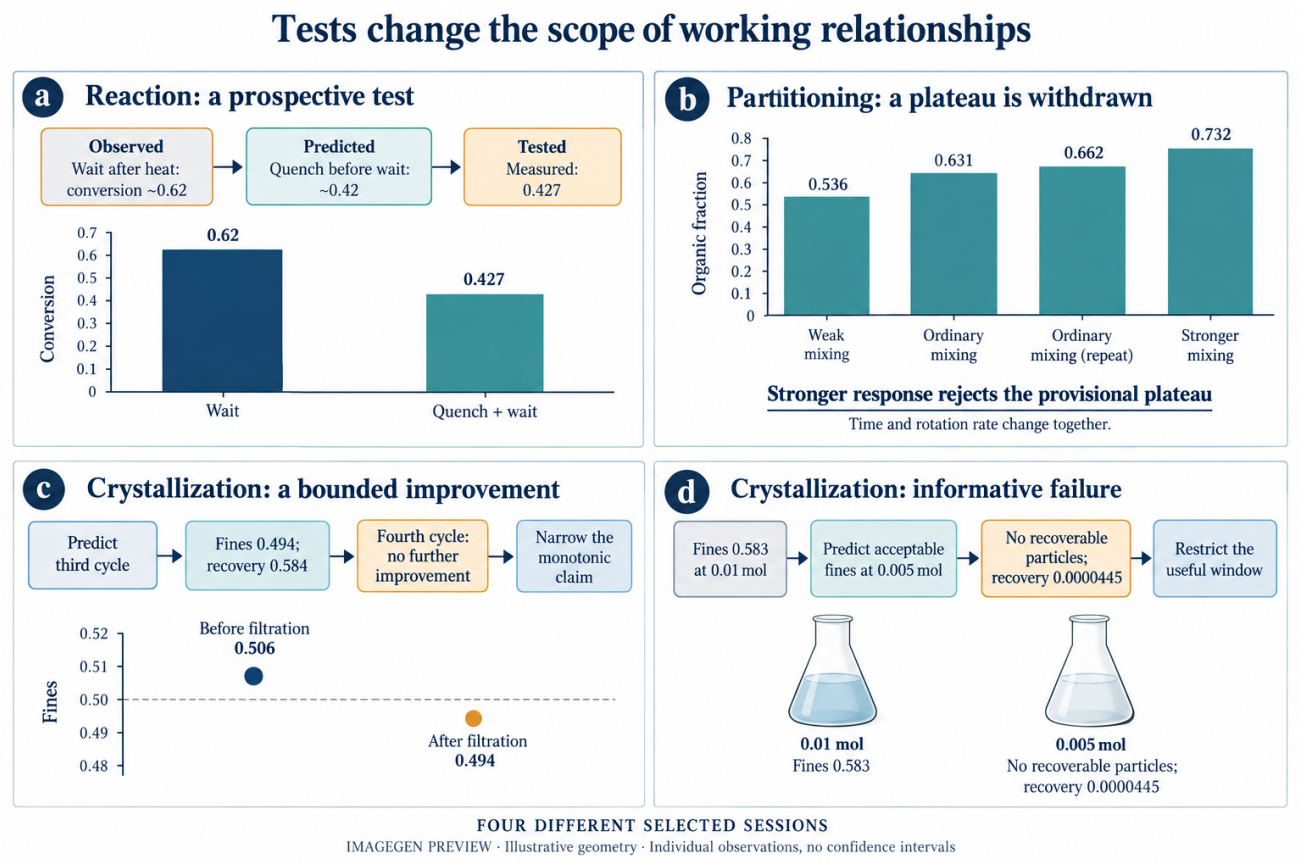

Extended Data Figure 5 | A local pattern and an unresolved explanation. a–c, Illustrative source-response patterns in equilibrium study A. Reagent and component commands add the same species in this execution version. d, The public history is condensed from an initial expectation through terminal uncertainty and later use; text is analytical summary, not quotations. e, Retained dilute dissociation forecasts, original 80% predictive intervals and withheld mean. Numerical accuracy does not establish a discovery mechanism. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.




Extended Data Figure 6 | Reconstructing what the experiment tested. a, Two rolled-back heating requests become a claimed 1,200 s negative result in one terminal reaction account; subsequent predictions invoke an induction or history effect. b, Another session excludes rolled-back heating from chemical interpretation. c, An intermediate crystallization note reports a 1 g seed test despite 0.05 g committed, then proposes a higher dose. Timing distinguishes later forecast use from an observed influence on subsequent source actions. The focal heating return combines outer success/completion labels with an explicit rollback error and zero time/temperature changes. This is joint evidence about interface presentation and reconstruction of experience. Separate histories are not randomized correction arms. ImageGen preview: text is analytical summary, geometry illustrative.




Extended Data Figure 7 | Different routes from evidence to prediction. a, Campaigns reaching at or below 1 mM among fifteen per model. b, An illustrative local dissociation plateau. c, The five-world Sol/Opaque dilute MAE is an aggregate, not a separate value for each world. d, Selected stages of one Astra campaign and its sealed forecast, with the original 80% predictive interval. Historical scientific accounts are retrospective. These observations do not establish fixed model traits. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.




Extended Data Figure 8 | Tests change the scope of working relationships. a, A prospective quench-before-wait prediction is followed by conversion 0.427, near the predicted 0.42. b, A stronger mixing condition leads to withdrawal of a provisional partitioning plateau; duration and rotation rate are confounded. c, A third crystallization cycle gives a near-threshold result, while a fourth fails to continue the improvement. d, Further dilution fails to preserve recoverable particles and narrows the proposed operating window. The panels are separate selected histories. Values are individual observations, not estimated treatment means or confidence intervals. ImageGen preview: geometry is illustrative; exact records accompany the manuscript.